Engineering the tools of scientific discovery

Engineering Design

ID - 23103-1

Hampton, VA

2026

Table of Contents

Problem Identification and Definition **3**

Information Gathering **4**

Possible Solutions **5-7**

#1: Smart Observation Station For Controlled Scientific Observation and Imaging **5**

#2: Pocket Field Science Sensor **6**

#3: Modular Smartphone Microscope Tester with Optical Alignment **7**

Chosen Solution: Modular Microscope **8-35**

Engineering Drawings and Operational Flow **19-31**

Engineering Design Process of the Creation of the Model **34-35**

Full Iteration Process **36-45**

Communication of Solution **46**

References **51-53**

Problem Identification and Definition

Access to scientific tools is a fundamental requirement for meaningful engagement in science education. Among these tools, the microscope is one of the most essential, enabling the observation of cells, microorganisms, and material structures that are otherwise invisible to the human eye. However, despite its importance in scientific discovery, access to functional microscopes remains limited in many educational environments. Traditional microscopes are often expensive, fragile, and require careful maintenance and training, which restricts their availability, particularly in under-resourced classrooms. Even when present, these devices are frequently shared among large groups of students, reducing hands-on interaction and limiting opportunities for inquiry-based learning. As a result, students are often confined to passive observation rather than active experimentation, which weakens conceptual understanding and reduces engagement in STEM disciplines. At the same time, modern smartphones have become nearly ubiquitous and are equipped with highly advanced imaging systems capable of capturing high-resolution data. These devices represent a powerful yet underutilized platform for scientific exploration. From an engineering perspective, smartphones already contain many of the core components of scientific imaging systems, including sensors, lenses, and computational processing capabilities. However, existing attempts to transform smartphones into scientific tools, particularly microscopes, are limited by significant design challenges. Current smartphone-based microscope solutions frequently suffer from poor optical alignment, image instability, inconsistent lighting, and a lack of structural support. These issues result in low-quality or unreliable images, making such systems ineffective for consistent educational use. The problem is therefore not simply a lack of microscopes but a lack of accessible, engineered systems that can transform widely available technology into reliable tools for scientific discovery. A successful solution must address several interdependent engineering challenges:

- **Optical Performance:** The system must provide sufficient magnification and clarity to resolve microscopic structures relevant to educational use.
- **Mechanical Stability:** The design must minimize vibration and maintain fixed alignment between optical components during operation.
- **Alignment Accuracy:** The system must ensure precise and repeatable positioning between the imaging device and optical elements.
- **Illumination Control:** Consistent and uniform lighting must be provided to enable clear image formation.
- **Usability:** The system must be intuitive and operable by users with little to no prior technical experience.
- **Cost and Accessibility:** The design must remain affordable and reproducible to allow widespread adoption in diverse educational settings.

These requirements highlight that the challenge is inherently a systems engineering problem, requiring the integration of optical, mechanical, and user-interface design into a single cohesive solution. Existing tools often optimize one aspect, such as cost or performance, at the expense of others, resulting in solutions that are either impractical or ineffective in real-world classrooms. Therefore, the problem can be defined as follows:

How can an engineered system be designed to transform a widely accessible device into a reliable, low-cost, and high-performance tool for scientific discovery, while maintaining optical clarity, structural stability, and ease of use in educational environments?

Addressing this problem directly supports the theme *"Engineering the Tools of Scientific Discovery"* by focusing not only on the use of scientific tools but also on the intentional design and optimization of those tools to expand access, improve functionality, and enhance how scientific investigation is performed. A successful solution would enable students to engage actively in observation, experimentation, and analysis, ultimately fostering deeper understanding, curiosity, and participation in STEM fields.

Information Gathering

Scientific progress has consistently been driven by the development of tools that extend human observation and measurement capabilities. From an engineering perspective, these tools are not simply accessories to discovery; they are foundational to it. One of the most influential examples is the invention of the microscope in the 17th century, which enabled scientists such as Robert Hooke and Antonie van Leuwenhoek to observe cells and microorganisms for the first time. This advancement fundamentally transformed biology, leading to the development of cell theory and modern microbiology. Historically, each major leap in scientific understanding has been closely tied to improvements in the tools available for investigation, demonstrating that innovation in engineering directly enables innovation in science. Despite this long-standing relationship between tools and discovery, access to scientific instrumentation remains uneven, particularly in educational environments. Many schools face limitations related to funding, equipment availability, and maintenance resources. As a result, students often lack opportunities to engage directly with scientific tools, relying instead on textbook-based or demonstration-based learning. Research indicates that reduced access to laboratory equipment limits students' ability to participate in inquiry-based learning, which is critical for developing problem-solving skills, analytical thinking, and scientific literacy. This creates a systemic gap in STEM education, where students are expected to understand complex scientific concepts without having the opportunity to explore them through direct experimentation.

The importance of addressing this challenge extends beyond the classroom. Access to hands-on scientific learning plays a key role in shaping future interest and participation in STEM fields. Students who are able to actively engage with scientific tools are more likely to develop curiosity, confidence, and a deeper understanding of scientific principles. In contrast, limited exposure can reduce engagement and discourage students from pursuing further study or careers in science and engineering. This has broader societal implications, as a lack of STEM participation can impact innovation, workforce development, and the ability to address global challenges that rely on scientific and technological advancement. In recent decades, the rapid global adoption of personal technology, particularly smartphones, has introduced a new dimension to this challenge. Smartphones are now among the most widely accessible forms of advanced technology, equipped with high-resolution sensors, imaging systems, and computational capabilities. From an engineering standpoint, this widespread availability presents a unique opportunity to expand access to scientific tools by leveraging existing technology rather than relying solely on traditional laboratory equipment. However, despite this potential, there remains a gap between the capabilities of these devices and their effective use in educational and scientific applications.

The challenge, therefore, is not only about creating new tools but about **engineering accessible, reliable, and effective systems that can broaden participation in scientific discovery**. Addressing this challenge has the potential to significantly impact education by:

- **Increasing accessibility:** Providing more students with the ability to engage in hands-on experimentation
- **Improving learning outcomes:** Strengthening understanding through active observation and inquiry
- **Enhancing engagement:** Encouraging curiosity and interest in STEM fields
- **Promoting equity in education:** Reducing disparities in access to scientific resources

From a broader perspective, developing solutions to this challenge supports the advancement of scientific literacy and innovation at a societal level. By expanding access to the tools of discovery, engineering can play a direct role in empowering the next generation of scientists, engineers, and problem-solvers.

Possible Solutions

Solution #1: Smart Observation Station for Controlled Scientific Observation and Imaging

The “Smart Observation Station” is designed to support controlled scientific observation by combining environmental isolation, magnification, and time-lapse imaging within a single compact system. Rather than relying on traditional open-air observation methods, this solution creates a stable and repeatable environment for studying biological processes such as plant growth, small organism behavior, or structural changes over time. By controlling lighting conditions and capturing continuous visual data, the system improves consistency in experimental observation and strengthens the connection between real-time phenomena and recorded analysis. Overall, the design emphasizes structured observation and improved data reliability in a classroom or lab setting.

The system is built around a sealed observation chamber that reduces interference from external light and environmental changes. Inside the chamber, a fixed and adjustable lighting system provides consistent illumination of the sample, while a mounted camera or imaging sensor with magnification capability captures detailed visual data over time. For optical-based experiments, light can be directed through controlled pathways such as narrow apertures or diffraction elements to observe interactions with materials under standardized conditions.

Key features include the following:

- Sealed chamber for environmental and lighting isolation
- Adjustable internal lighting for consistent illumination
- Mounted magnified imaging system for detailed observation
- Controlled optical pathways for light-based experiments
- Time-lapse imaging for long-term data capture
- Modular components for interchangeable experimental setups

From an engineering perspective, the design prioritizes alignment stability, modularity, and environmental control. Fixed mounts reduce positional error between trials, while the enclosed structure ensures external conditions do not distort results. A time-lapse system automates image capture at set intervals, allowing long-term processes to be recorded without constant user input. Optical co-nfigurations remain consistent through controlled alignment of light sources, samples, and imaging systems, improving repeatability across experiments.

Despite its structured design, the system has several limitations. In optical applications, much of the analysis depends on interpreting captured patterns rather than direct observation, which can increase cognitive demand. In biological experiments, short or rapid changes may be missed between imaging intervals. Additionally, calibration of lighting, focus, and alignment is required for accurate results, adding setup complexity.

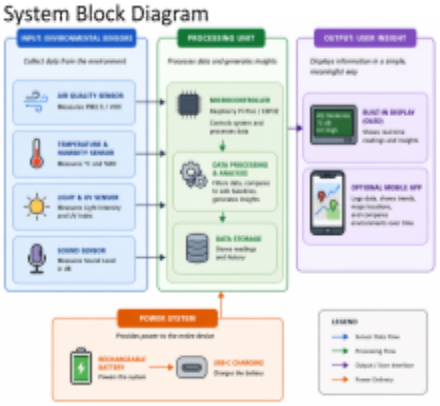
Overall, the Smart Observation Station provides a highly controlled and consistent approach to scientific observation. Its strengths lie in improved repeatability, long-term recording, and environmental stability, making it suitable for classroom and laboratory use. However, its reliance on controlled conditions and indirect interpretation reduces its flexibility, making it less effective for quick or highly intuitive experimentation in uncontrolled environments.

Solution #2: Pocket Field Science Sensor

The Pocket Field Science Sensor (PFSS) is designed to extend scientific investigation beyond the classroom by enabling real-time environmental data collection in everyday settings. Rather than relying on stationary laboratory equipment, this solution allows students to measure and interpret conditions such as air quality, temperature, humidity, light exposure, and sound levels directly within their surroundings. By shifting scientific observation into real-world environments, the system promotes continuous, inquiry-based learning and strengthens the connection between theoretical concepts and practical application. The PFSS is constructed as a compact embedded system centered around a low-power microcontroller, such as the Raspberry Pi Pico, integrated with multiple environmental sensors. These sensors collect data on key environmental variables, which are then processed in real time by the microcontroller. To improve usability, the system converts raw numerical data into simplified, interpretable outputs, allowing users to quickly understand environmental conditions without requiring advanced technical knowledge. This approach prioritizes accessibility and reduces the complexity typically associated with scientific data analysis.

Figure 1

System Block Diagram



The engineering design emphasizes portability, energy efficiency, and ease of use. The device is battery-powered and housed in a compact, durable enclosure, allowing it to be used in a wide range of environments. By integrating multiple sensors into a single system and processing data in real time, the PFSS provides immediate feedback to the user. Basic sensor fusion techniques are applied to combine multiple inputs into more meaningful insights, improving the overall interpretability of the data while maintaining system simplicity.

Despite its accessibility, the PFSS introduces several limitations that impact its effectiveness as a scientific tool. The use of low-cost sensors reduces measurement accuracy compared to laboratory-grade instruments, and environmental factors such as interference or changing conditions can affect data reliability. Additionally, the simplification of outputs, while beneficial for usability, limits the depth of analysis that can be performed. These constraints represent a trade-off between ease of use and scientific precision. Overall, the PFSS provides a practical and engaging approach to environmental data collection by making scientific observation portable and accessible. Its strengths lie in mobility, real-time feedback, and ease of use; however, limitations in accuracy, control, and analytical depth reduce its suitability for controlled or high-precision scientific investigation, making it less viable as a primary solution. Despite these limitations, the PFSS demonstrates how environmental sensing can be made more accessible through compact and low-cost engineering design. It effectively lowers the barrier to entry for students who may not have access to traditional laboratory equipment. The system also highlights the importance of simplifying complex data without completely removing its scientific value. From an engineering perspective, it illustrates how multi-sensor integration can be used to create meaningful real-world insights. However, the trade-off between simplicity and precision remains a key constraint that limits its application in rigorous experimental settings. Ultimately, while the PFSS is valuable as an educational engagement tool, it does not fully meet the performance requirements needed for advanced scientific analysis.

Solution #3: Modular Smartphone Microscope Tester with Optical Alignment

The final solution selected for development is a modular smartphone-integrated microscope system with engineered optical alignment. This design directly addresses the lack of accessible scientific tools by providing a low-cost, portable, and easy-to-use system capable of producing clear visual observations. Unlike other proposed solutions that rely on indirect data or limited applications, this system enables direct observation, making it more intuitive and effective for both educational and field use. By leveraging the high-resolution camera of a smartphone, the design eliminates the need for expensive standalone imaging equipment while maintaining functional performance.

A key strength of this solution is its modularity, allowing the system to adapt to multiple experiments rather than serving a single purpose. Users can reconfigure components based on their needs, supporting a wide range of scientific investigations.

- Interchangeable lenses (4×, 10×, 40×) provide multiple magnification levels
- Adjustable illumination improves clarity across different specimens
- Smartphone integration enables real-time imaging and data capture
- Replaceable components increase durability and reduce long-term cost

From an engineering standpoint, the system is designed to balance performance with accessibility. Traditional microscopes rely on expensive precision components, whereas this design uses simplified structures and widely available materials to achieve similar functionality at a lower cost.

- Compact and lightweight for portability
- Low-cost materials enable scalability
- Simplified mechanisms reduce user error and setup time

The system is also safe and easy to use, making it suitable for a wide range of users without specialized training.

- No hazardous materials or high-force components
- Minimal maintenance required
- Intuitive setup for beginners

This solution supports hands-on experimentation by allowing users to adjust variables and immediately observe results, reinforcing scientific and engineering concepts through active learning. Its applications extend across multiple groups, including students, educators, field scientists, and general users, making it significantly more versatile than alternative solutions. Additionally, the system encourages iterative problem-solving, as users can continuously refine their setup to improve image quality and experimental outcomes. The integration of digital imaging also allows observations to be recorded, shared, and analyzed, strengthening both scientific communication and collaboration. Over time, this scalability and adaptability make the system suitable for a wide range of environments, from classrooms to fieldwork.

Compared to the other concepts, the modular microscope provides the strongest balance of usability, versatility, and educational value. It delivers immediate visual feedback, supports multiple disciplines, and remains cost-effective and scalable. Ultimately, this solution fulfills the goal of engineering the tools of scientific discovery by expanding access to meaningful scientific observation and enabling a broader range of users to actively engage in science.

Most Plausible Solution

To determine the most effective solution, all three proposed concepts were evaluated using key engineering criteria: cost, safety, usability, versatility, scalability, and educational value. Each solution was analyzed based on its ability to meet the project objective of creating an accessible and effective STEM tool.

Solution Comparison Table

Criteria	Solution 1: Smart Scientific Observation Station	Solution 2: Pocket Field Science Sensor	Solution 3: Modular Microscope
Cost	Moderate (~$75–$150 due to enclosure, camera system, and lighting components)	Low (~$30–$60 depending on sensor selection)	Low (<$30 target)
Safety	Safe but requires controlled setup and proper handling of enclosed system	Very safe; low-risk electronic device	Safe for classroom and field use; no hazardous components
Ease of Use	Moderate; requires setup, calibration, and understanding of system controls	Easy: simplified outputs reduce complexity	Simple and intuitive; minimal setup with guided alignment
Versatility	Moderate; supports biological and some optical experiments but limited flexibility	Moderate; focused mainly on environmental sensing	High; supports biology, materials science, optics, and field observation
Portability	Low to moderate; enclosed chamber limits mobility	High; compact and handheld	High; lightweight and smartphone-integrated
Maintenance	Moderate; requires calibration and component alignment	Low; minimal upkeep of sensors	Low; simple cleaning and replaceable modular components. The majority are 3D printable
Educational Impact	High for structured experiments but limited for open exploration	Moderate; good for engagement but limited depth	High; supports hands-on, inquiry-based, and multi-disciplinary learning
Advantages	Controlled environment, consistent results, time-lapse capability	Portable, real-time data; easy to use; encourages field exploration	Direct observation, highly versatile, low cost, portable, modular, strong engagement
Disadvantages	Bulky, less intuitive, requires calibration, limited flexibility	Lower accuracy, limited depth of analysis, indirect data interpretation	Minor optical limitations and alignment sensitivity, but acceptable trade-offs

Final Selection Justification

Based on this evaluation, the modular smartphone microscope (Solution 3) was selected as the most plausible solution. Unlike the other concepts, it achieves a strong balance across all criteria while directly addressing the identified problem. It provides a safe, low-cost, portable, and highly versatile platform that enables hands-on experimentation across multiple STEM disciplines. This decision was supported by both theoretical comparison and practical feasibility, ensuring the selected solution is not only effective but also scalable for real-world educational use.

Prototype Overview and Engineering Objective

The selected solution, a modular, smartphone-integrated microscope system, was developed to directly address the lack of accessible, affordable, and functionally reliable microscopy tools in educational environments. The primary engineering objective was not simply to "create a microscope," but to design a system that balances optical performance, mechanical stability, usability, and cost constraints simultaneously. Traditional microscopes achieve high performance through precision-manufactured optical assemblies, rigid metal frames, and calibrated mechanical components. However, these systems are often cost-prohibitive and not scalable for widespread classroom use. In contrast, this prototype leverages the advanced imaging capabilities of modern smartphones and enhances them through external engineered subsystems that replicate the essential functions of a traditional microscope at a fraction of the cost. The design objectives were explicitly defined as:

- **Optical Performance:** Achieve sufficient magnification and clarity to observe cellular-scale structures
- **Mechanical Stability:** Eliminate vibration and maintain fixed optical alignment during use
- **Ease of Use:** Ensure intuitive operation for users with no prior microscopy experience
- **Modularity:** Allow interchangeable components (lenses, lighting, mounts)
- **Cost Efficiency:** Maintain a target cost below $25 for scalability

Each of these objectives introduced constraints that influenced the design. For example, increasing magnification typically reduces depth of field and increases sensitivity to vibration, which requires corresponding improvements in structural stability and focus control. As a result, the final prototype is best understood as a system-level optimization, where multiple engineering trade-offs were balanced to achieve the most practical and effective outcome.

Implementation and Integration Plan

The implementation of the modular smartphone microscope system is not limited to the construction of a working prototype; rather, it represents a complete strategy for deploying a scalable, accessible, and impactful scientific tool. This section outlines the reasoning behind selecting this solution, the populations it serves, how it is used in practice, and how it can be effectively integrated into real-world environments. The goal is to demonstrate that the design is not only technically functional but also practical, adoptable, and meaningful in advancing scientific discovery.

Justification for Selection and Real-World Relevance

The modular smartphone microscope was selected as the final solution because it achieves a balance between engineering performance and real-world feasibility that the alternative solutions could not. While other concepts addressed specific aspects of scientific learning, this system directly enables visual, hands on interaction with microscopic phenomena, which is one of the most fundamental experiences in science education. This solution is particularly effective because it leverages an existing, widely available technology platform, the smartphone, and

enhances it through engineered subsystems rather than replacing it. This significantly reduces barriers to implementation. Key reasons for selection include the following:

- **Direct observation capability:**
    - Unlike indirect measurement systems (e.g., sensors or spectroscopy), this device allows users to *see* structures such as cells and materials in real time, making scientific concepts more tangible and intuitive.
- **Balanced engineering performance:**
    - The system achieves sufficient magnification, clarity, and stability while remaining low-cost, which is critical for classroom adoption.
- **Accessibility and scalability:**
    - By using common materials and smartphone integration, the system can be implemented in a wide range of environments without requiring specialized infrastructure.
- **Alignment with the project theme:**
    - The design directly embodies *engineering the tools of scientific discovery* by transforming a consumer device into a functional scientific instrument.

Overall, the selection was based not only on technical capability but also on the system's ability to create meaningful, repeatable, and accessible scientific experiences.

Target Users and Beneficiaries

A major strength of this solution is its broad applicability. The system was intentionally designed to benefit multiple user groups, each of which interacts with the device in a slightly different way depending on their needs. Primary user groups and their benefits:

- **Students (Elementary School through College):**
    - Students benefit from the ability to directly observe microscopic structures, which reinforces theoretical knowledge through visual experience. This improves retention and understanding of key concepts such as cell structure, material properties, and optical behavior. Additionally, hands-on interaction promotes curiosity and encourages independent exploration.
- **Educators and Classrooms:**
    - Teachers benefit from a tool that is easy to implement, requires minimal setup, and does not depend on expensive laboratory equipment. The system allows for:
    - Increased student participation (more students can use individual devices)
    - More efficient lesson integration (reduced setup and calibration time)
    - Greater flexibility in experiment design
- **Field Scientists and Resource-Limited Researchers:**
    - In environments where traditional microscopes are impractical, this system provides a portable alternative for preliminary observations. While not a replacement for laboratory-grade equipment, it serves as an effective tool for quick analysis and documentation.
- **Independent Learners and Hobbyists:**
    - The low cost and simplicity of the system make it accessible to individuals outside formal education, supporting self-directed learning and experimentation.
- **Underserved and Low-Resource Communities:**

    - By reducing cost and complexity, the system enables broader participation in scientific learning where access to equipment is limited.

This wide range of beneficiaries demonstrates that the solution is not niche; it is designed for maximum impact across educational and practical contexts.

Practical Use and Operational Workflow

The system was engineered to ensure that operation is intuitive while still maintaining precise control over critical variables such as alignment and focus. The workflow reflects a balance between simplicity and technical accuracy. Operational process:

- The smartphone is placed into the alignment mount and positioned using the guided rail system
- The user selects the desired magnification level via the lens carousel
- The specimen is placed on the stage and illuminated using the integrated LED system
- The focus mechanism is adjusted to achieve a sharp image
- The smartphone captures and displays the magnified image in real time

While this process appears simple, each step is supported by engineered constraints that improve reliability:

- **Guided alignment system:** prevents misalignment by restricting motion to controlled axes
- **Fixed lens positioning:** ensures correct working distance without manual calibration
- **Stable stage design:** minimizes vibration during focusing

These design choices transform what is typically a trial-and-error process into a controlled and repeatable system, significantly improving usability for beginners.

Integration into Educational and Real-World Environments

For a solution to be effective, it must integrate seamlessly into the environments in which it is used. This system was specifically designed to require minimal external support, allowing for flexible deployment.

Classroom integration:

- Can be incorporated into biology, physics, and engineering curricula
- Supports both structured labs and open-ended exploration
- Enables small-group or individual use due to low cost

Field and outdoor use:

- Compact and lightweight design allows easy transport
- Does not require external lab equipment or a complex setup
- Enables real-time observation in natural environments

Digital and hybrid learning environments:

- Captured images can be saved, shared, and analyzed
- Supports remote learning and independent experimentation

- Integrates easily into digital assignments and reports

The system's adaptability ensures that it is not limited to a single use case, increasing its effectiveness and long-term value.

Scalability and Manufacturing Feasibility

A key aspect of implementation is the ability to reproduce the system efficiently and affordably. The design was developed with scalability as a primary consideration.

Factors enabling large-scale implementation:

- **3D-printed components:**
  Allow rapid production and easy modification without specialized manufacturing equipment
- **Accessible materials:**
  Components such as PLA, acrylic, LEDs, and standard lenses are widely available and low-cost
- **Simple assembly process:**
  The modular design allows parts to be assembled with minimal technical expertise
- **Repairability and replacement:**
  Individual components can be replaced without rebuilding the entire system

These characteristics ensure that the system can be produced not only in industrial settings but also in schools, makerspaces, and community labs, significantly increasing accessibility.

Educational and Societal Impact

The impact of this solution extends beyond technical performance and directly contributes to improving how users interact with science.

Key impacts include:

- Increased engagement through hands-on experimentation
- Improved conceptual understanding through visual learning
- Development of critical thinking and observational skills
- Greater accessibility to scientific tools across diverse populations

By enabling users to actively participate in scientific observation, the system shifts learning from passive to interactive, aligning with modern educational approaches that emphasize inquiry-based learning.

Long-Term Adaptability and Future Expansion

The modular nature of the system ensures that it can evolve rather than become obsolete.

Potential future improvements:

- Integration of higher-quality optical components for improved resolution
- Addition of digital measurement or image analysis tools

- Enhanced stabilization or automated focusing mechanisms
- Expanded compatibility with different smartphone designs

Because each subsystem operates independently, improvements can be implemented without redesigning the entire system. This reflects strong engineering practice, where flexibility and adaptability are built into the design.

Conclusion of Implementation Strategy

The modular smartphone microscope is not only a technically effective solution but also a highly implementable one. Its ability to integrate into classrooms, fieldwork, and independent learning environments, combined with its scalability and accessibility, demonstrates that it is a practical and impactful tool for expanding access to scientific discovery. By addressing both engineering constraints and real-world application requirements, the system represents a complete and well-justified solution capable of meaningful adoption.

System Architecture and Functional Integration

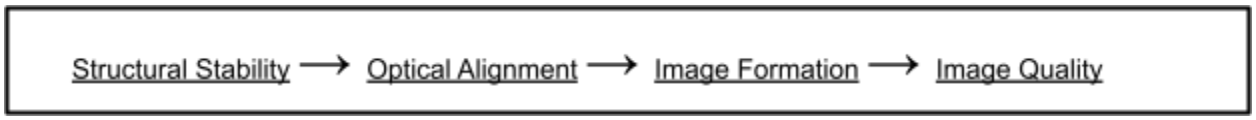
The microscope was designed as a multi-subsystem integrated system, consisting of:

- Optical Subsystem
- Structural Frame
- Smartphone Alignment Mechanism
- Specimen Stage and Focus Control
- Illumination System

Rather than functioning independently, these subsystems operate as a coupled system, where the performance of each directly impacts the others. This interdependence is critical to understanding the engineering approach. For example:

- Misalignment in the smartphone mount directly reduces optical clarity
- Poor illumination increases sensor noise, reducing effective resolution
- Structural instability introduces motion blur at high magnification

This relationship can be understood as a dependency chain:



To ensure reliable performance, the design process focused on minimizing variability at each stage of this chain.

Optical Subsystem Design and Performance

The optical subsystem is the most critical component of the system, as it determines magnification, resolution, and image clarity. The design utilizes a rotating objective lens carousel, containing 4×, 10×, and 40× lenses, allowing users to switch magnification levels without realigning the microscope. This design choice was driven by the need to balance flexibility and repeatability:

- Fixed single-lens systems limit usability
- Fully adjustable systems increase complexity and alignment error
- A click-stop carousel provides both modularity and precision

Key Optical Design Considerations:

- Working Distance Control:
    - Each lens must be positioned at a specific distance from the specimen to produce a focused image. The carousel ensures this distance is fixed and repeatable.
- Optical Axis Alignment:
    - The cylindrical housing constrains the lens position to maintain alignment with the smartphone camera sensor, minimizing distortion.
- Field of View Optimization:
    - Only the central region of each lens is used, where optical quality is highest, to reduce distortion and blurring at the edges.

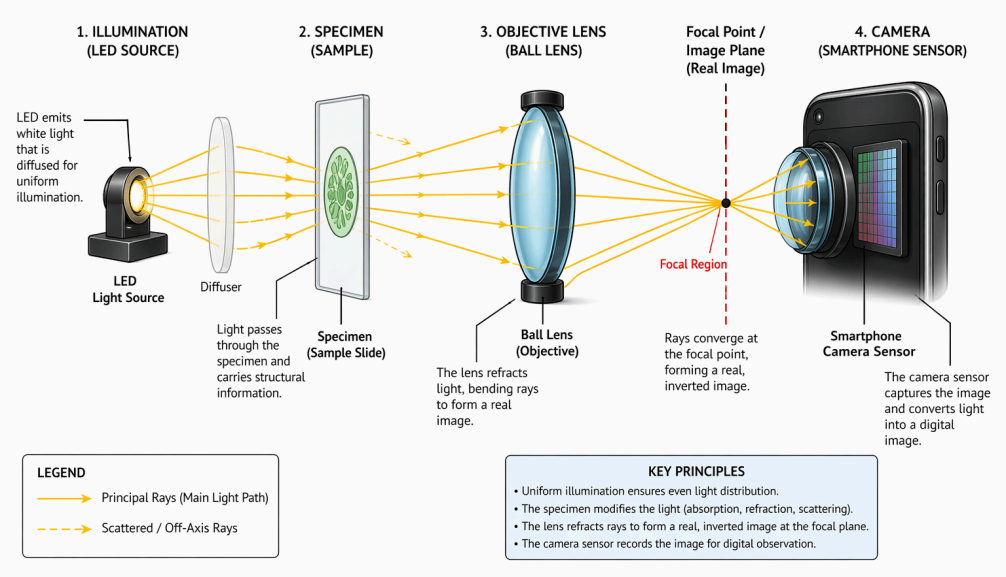
Engineering Trade-offs:

- Higher magnification (40×) provides more detail, but:
    - Reduces brightness
    - Decreases the depth of the field
    - Increases sensitivity to vibration
- Lower magnification (4×, 10×):
    - Provides a larger field of view
    - Easier focusing
    - More stable imaging

This multi-lens approach allows the system to adapt to different use cases rather than forcing a single compromise.

Figure 2

Optical Path Diagram (Ray diagram showing LED → specimen → lens → camera)



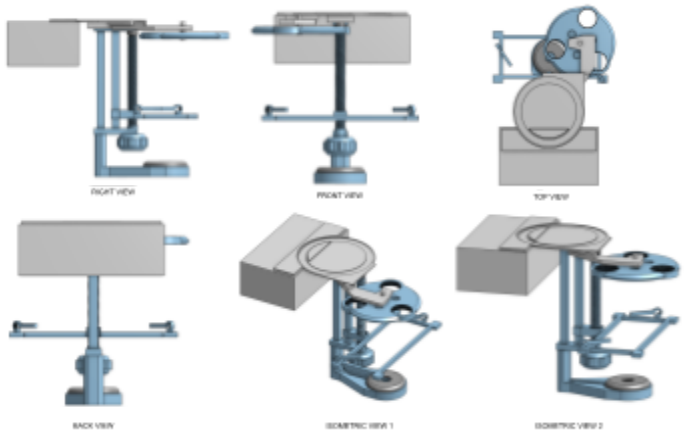
Note. This figure illustrates the optical path of a smartphone-based microscope, showing how light travels from the LED source through the specimen and into the objective lens. The lens refracts and converges the light rays to form a real, inverted image at the focal plane. This image is then captured by the smartphone camera sensor, which converts the light into a digital signal. Uniform illumination and precise alignment are essential for achieving a clear and accurate image. Created using CanvaPro.

Structural Frame and Stability Engineering

The structural subsystem was engineered to ensure that the optical and alignment systems remain fixed relative to one another during operation. At higher magnifications, the microscope becomes increasingly sensitive to mechanical disturbances, meaning that even minimal vibrations or positional shifts can significantly degrade image quality. As a result, structural stability was treated not as a secondary consideration but as a primary design requirement directly tied to optical performance. The frame consists of a rigid base platform, vertically oriented support columns, and reinforced connection points that collectively define the geometry of the system.

Figure 3

Complete System Assembly Diagram



As shown in Figure 3 (Complete System Assembly Diagram), the vertical supports maintain a fixed distance between the smartphone mount and specimen stage, which is critical for preserving focal alignment across different magnification levels. Any deviation in this spacing would result in defocusing or inconsistent imaging. To address these challenges, several key engineering strategies were implemented:

- **Support Leg:**
    - The support leg was selected to increase stability and reduce the likelihood of tipping during interaction. The support leg aids by standing as a verticle support that extends to the same length of the entire module.
- **Rigid Material Selection:**
    - PLA was used for 3D-printed components due to its sufficient stiffness and ease of fabrication. While not as strong as metal, it provides an effective balance between rigidity and cost for educational applications.
- **Minimized Degrees of Freedom:**
    - The design intentionally limits unnecessary movement by constraining components to fixed paths. This reduces the risk of unintended displacement during use.
- **Reinforced Joints and Load Paths:**
    - Structural connections (strategically placed infill patterns) were designed to prevent flexing under load, ensuring that forces applied during focusing or adjustment do not propagate as vibrations through the system.

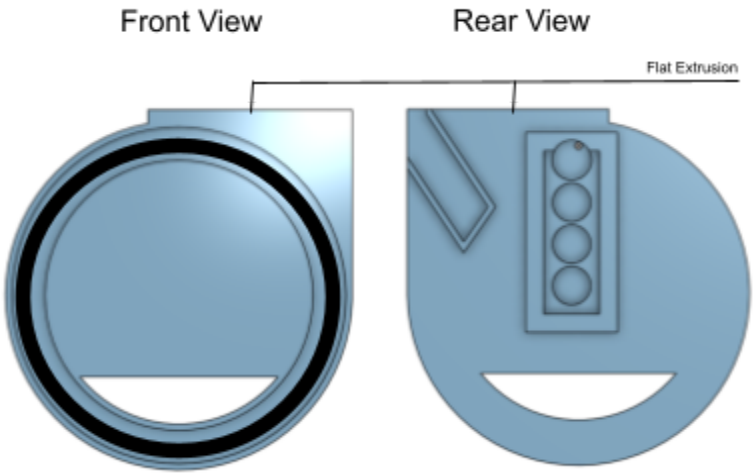
The importance of these features becomes evident when considering magnification effects. At approximately 40× magnification, a physical displacement of even 0.1 mm can translate into a large apparent shift in the observed image. Therefore, maintaining structural rigidity is essential to ensuring that the optical subsystem can perform as intended.

Smartphone Mounting and Optical Alignment Mechanism

The smartphone mounting and alignment subsystem was developed to address one of the most critical failure points in smartphone-based microscopy: precise and repeatable alignment between the camera sensor and the external optical system. Without proper alignment, the system cannot produce usable images, regardless of lens quality or lighting conditions. We implemented a magnetic mounting alignment system, ensuring that the phone is able to be repeatedly secured into the mount. The attachment mechanism uses a ring embedded into the mount housing, while a flat extrusion aligns the main housing perfectly parallel to the camera bump.

Figure 4

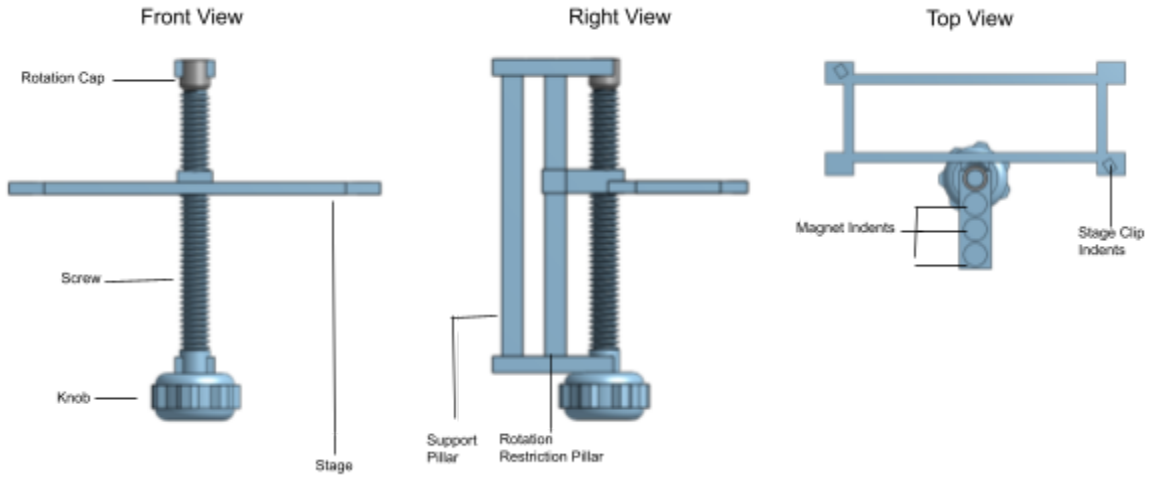
Smartphone Magnetic Mounting Alignment Mechanism



Note. The black ring shown in image represents the magnetic ring used for the magnetic mounting. Screenshot taken of the cad model.

Figure 5

Vertical Smartphone Alignment Mechanism



Note. The vertical smartphone alignment mechanism can unattach from the mount housing allowing the microscope to be used without the stage.

As illustrated in Figure 4 and 5, the mounting system incorporates a guided sliding rail that allows controlled vertical adjustment of the smartphone relative to the lens. This enables the user to align the camera lens directly with the optical axis of the selected objective lens. The design solves multiple alignment challenges simultaneously:

- **Controlled Adjustment Instead of Free Movement:**
    - The sliding rail restricts motion to a single axis, preventing random misalignment due to rotation, while still allowing fine positioning.
- **Repeatable Positioning Through Locking Mechanism:**
    - Once alignment is achieved, the magnet secures the phone in place. This eliminates the need for repeated calibration and ensures consistent imaging across multiple uses.

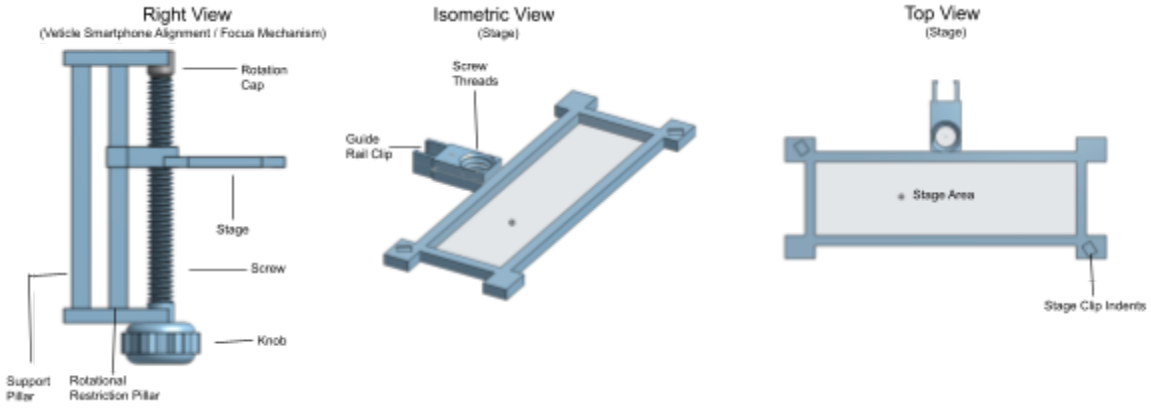
This ensures alignment is controlled by the device itself rather than depending on the user to position it correctly. This significantly improves reliability, reduces setup time, and ensures that users with minimal experience can achieve proper alignment without trial-and-error.

Specimen Stage and Focus Control System

The specimen stage and focus control subsystem are responsible for maintaining the correct distance between the specimen and the optical lens, which directly determines image sharpness.

Figure 6

Specimen Stage and Focus Mechanism



As shown in Figure 6 (Specimen Stage and Focus Mechanism), the stage incorporates a guided vertical adjustment system that allows incremental movement of the specimen relative to the lens. This design enables users to fine-tune focus while maintaining system stability. The engineering requirements for this subsystem were defined by the following constraints:

- The system must allow fine vertical adjustments without sudden movement
- The specimen must remain centered and stable during focusing
- Adjustments must not introduce vibration or misalignment

To meet these requirements, the design includes:

- **Flat Stage Platform:**
    - Provides a stable surface for slides and prevents tilting.
- **Slide Positioning Guides:**
    - Ensure consistent placement of specimens directly under the optical axis.
- **Controlled Vertical Motion:**
    - Allows users to adjust focus in small increments, reducing overshoot and improving precision.

The shallow depth of field means that only a thin cross-section of the specimen is in focus at any given time. While this increases the difficulty of focusing, it also enhances the system's ability to resolve fine structural details. The focus mechanism, therefore, plays a critical role in balancing usability with optical performance.

Illumination Subsystem and Light Optimization

The illumination subsystem was designed to provide consistent and uniform lighting conditions, which are essential for producing clear and high-contrast images. In microscopy, insufficient or uneven lighting can reduce visibility, introduce noise, and obscure fine details. The system utilizes a transmitted light configuration, in which a low-voltage LED is positioned beneath the specimen. As light passes through the sample, it carries structural information that is captured by the optical system and recorded by the smartphone camera. As illustrated in Figure 3 (Optical Path Diagram), this configuration ensures that the specimen is evenly illuminated from below. To improve lighting quality, a diffuser layer was incorporated to distribute light uniformly across the specimen. This reduces harsh shadows and prevents localized overexposure, both of which can negatively impact image clarity. Key design considerations for the illumination subsystem include:

- **Uniform Light Distribution:**
    - Achieved through the use of a diffuser to eliminate hotspots.
- **Sufficient Light Intensity:**
    - Ensures that the camera sensor receives enough light to minimize noise.
- **Energy Efficiency and Accessibility:**
    - Low-power LEDs were selected to maintain affordability and ease of use.

The interaction between illumination and camera performance is critical. Because the system relies on a smartphone sensor, which performs best under adequate lighting, optimizing illumination directly improves image quality without increasing system complexity or cost.

Dimensional Specifications and Engineering Parameters

To ensure consistent performance, the prototype was designed within a defined set of dimensional and operational parameters. These values were not arbitrary, but rather selected to balance optical performance, usability, and mechanical stability. The parameters define the operational envelope of the system. For example, maintaining the working distance within a narrow range is essential for achieving proper focus, while alignment tolerances ensure that the optical axis remains centered on the camera sensor. From an engineering standpoint, these constraints serve two purposes:

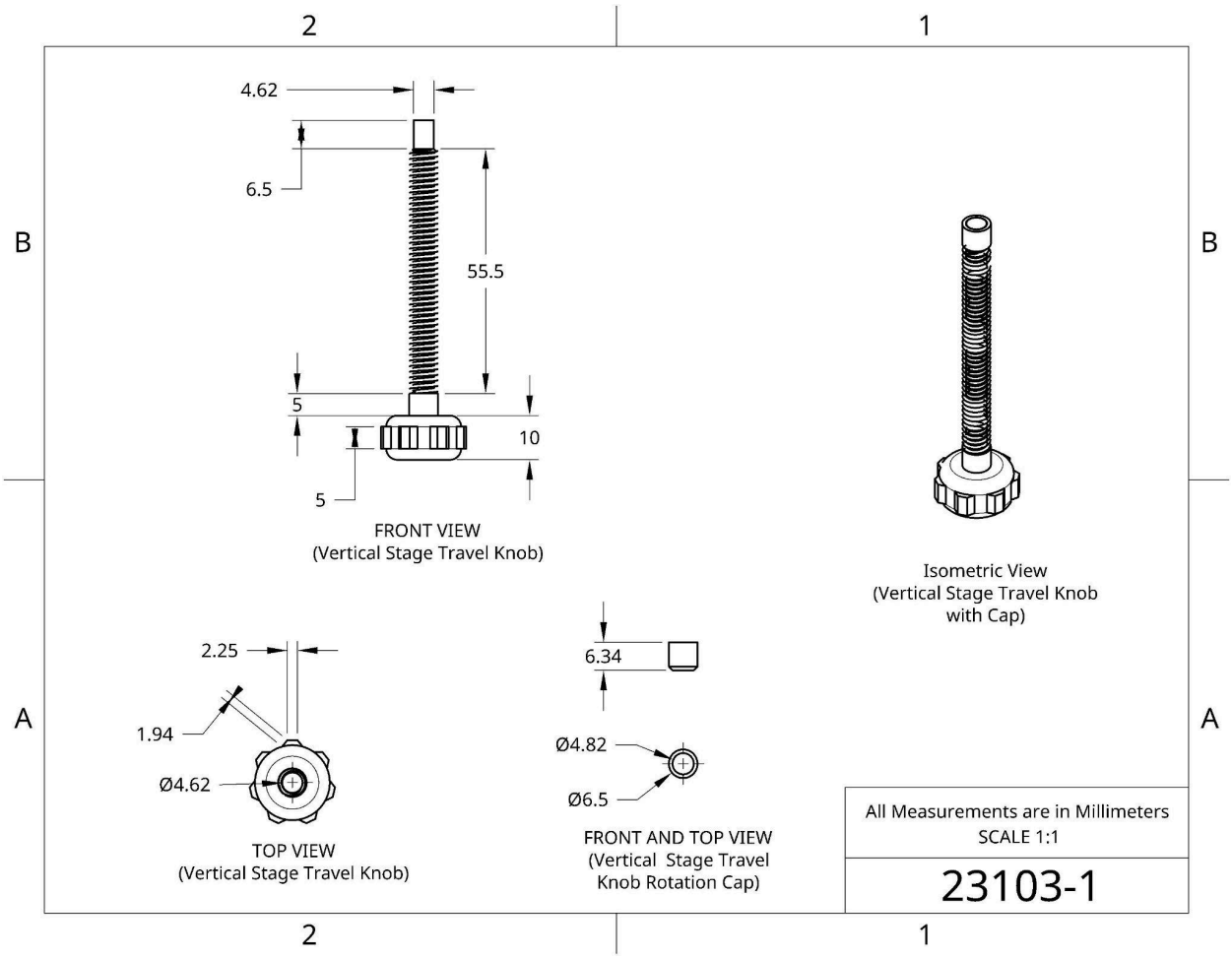
- They ensure repeatability, allowing consistent results across multiple uses
- They limit variability, reducing the impact of user error on system performance

By explicitly defining these parameters, the design transitions from a conceptual model to a controlled and measurable engineering system.

Engineering Drawings and Operational Flow

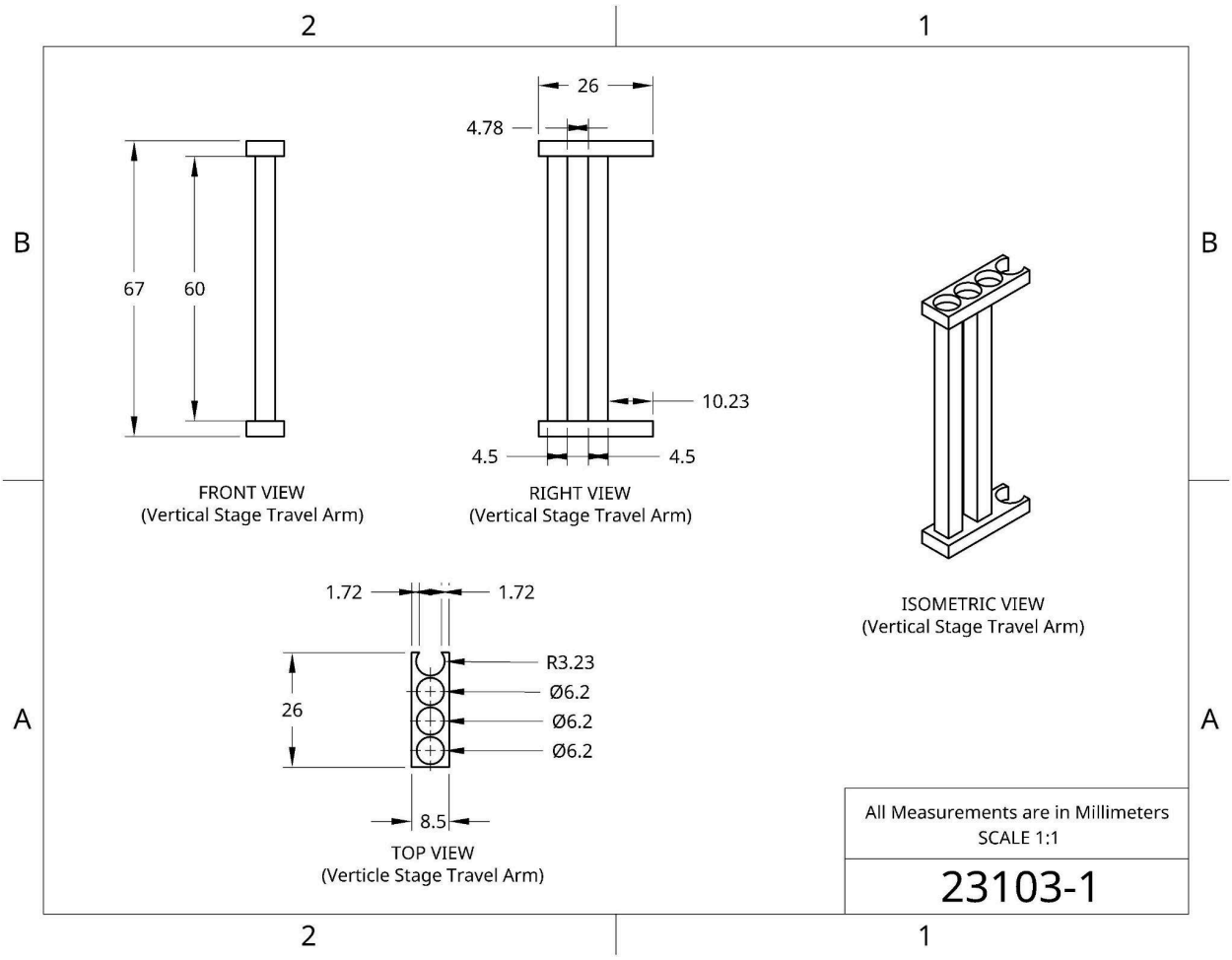
To fully communicate the design and functionality of the prototype, multiple visual representations were developed.

Figure 7: Vertical Stage Travel Knob



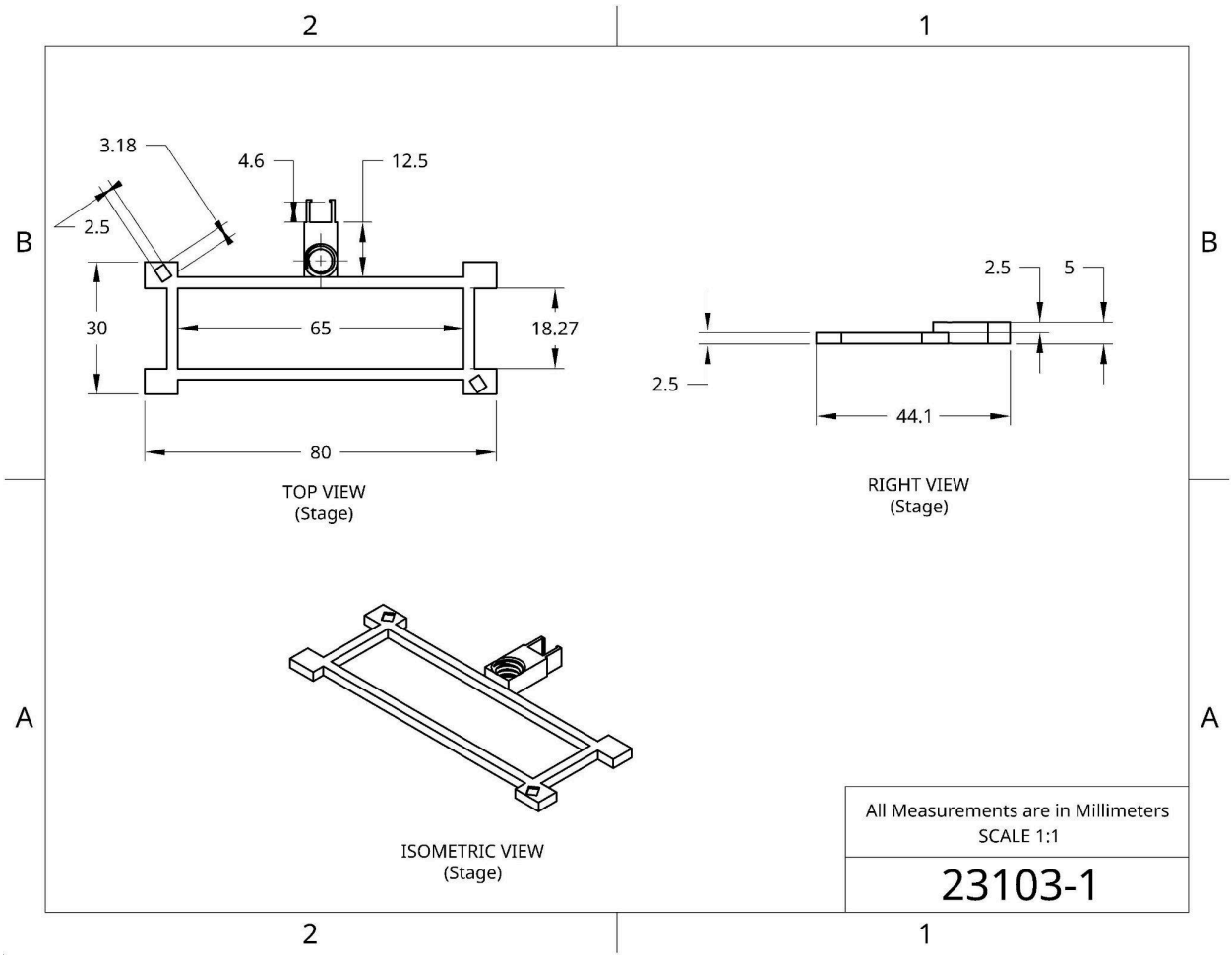
Note. The vertical stage travel knob serves as the primary user interface for fine focus adjustment. Its design features chamfered ridges around the perimeter to provide an ergonomic tactile grip for precise manipulation. The assembly utilizes a threaded shaft to translate rotational movement into linear vertical travel. To ensure smooth operation, the top of the knob includes a through-hole that allows the rotating sleeve from the stage travel arm to seat securely, maintaining concentricity and preventing "wobble" during use.

Figure 8: Vertical Stage Travel Arm



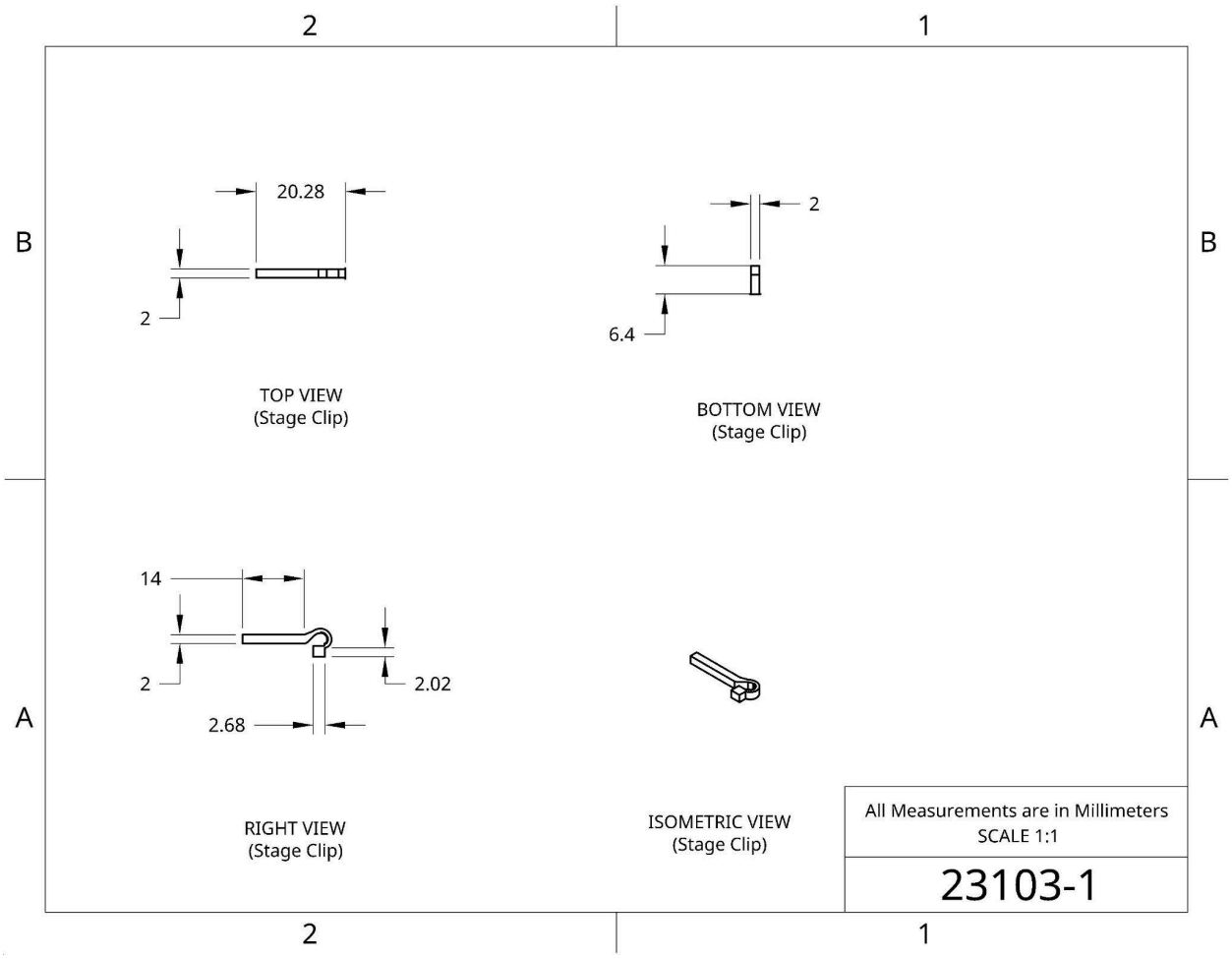
Note. Acting as the structural backbone of the focusing travel mechanism, the travel arm maintains a rigid reference for vertical motion. It consists of a rear main support pillar and a front square guide pillar. This dual-pillar configuration is critical: the square cross-section of the guide pillar constrains the stage, preventing unwanted rotation and ensuring the specimen remains centered on the optical axis. The assembly is anchored by a solid bottom clip, while the top clip houses the rotating sleeve for the focus knob. By distributing loads across four contact surfaces, the arm eliminates mechanical flex and maintains a fixed distance between the stage and the lens carousel.

Figure 9: Stage



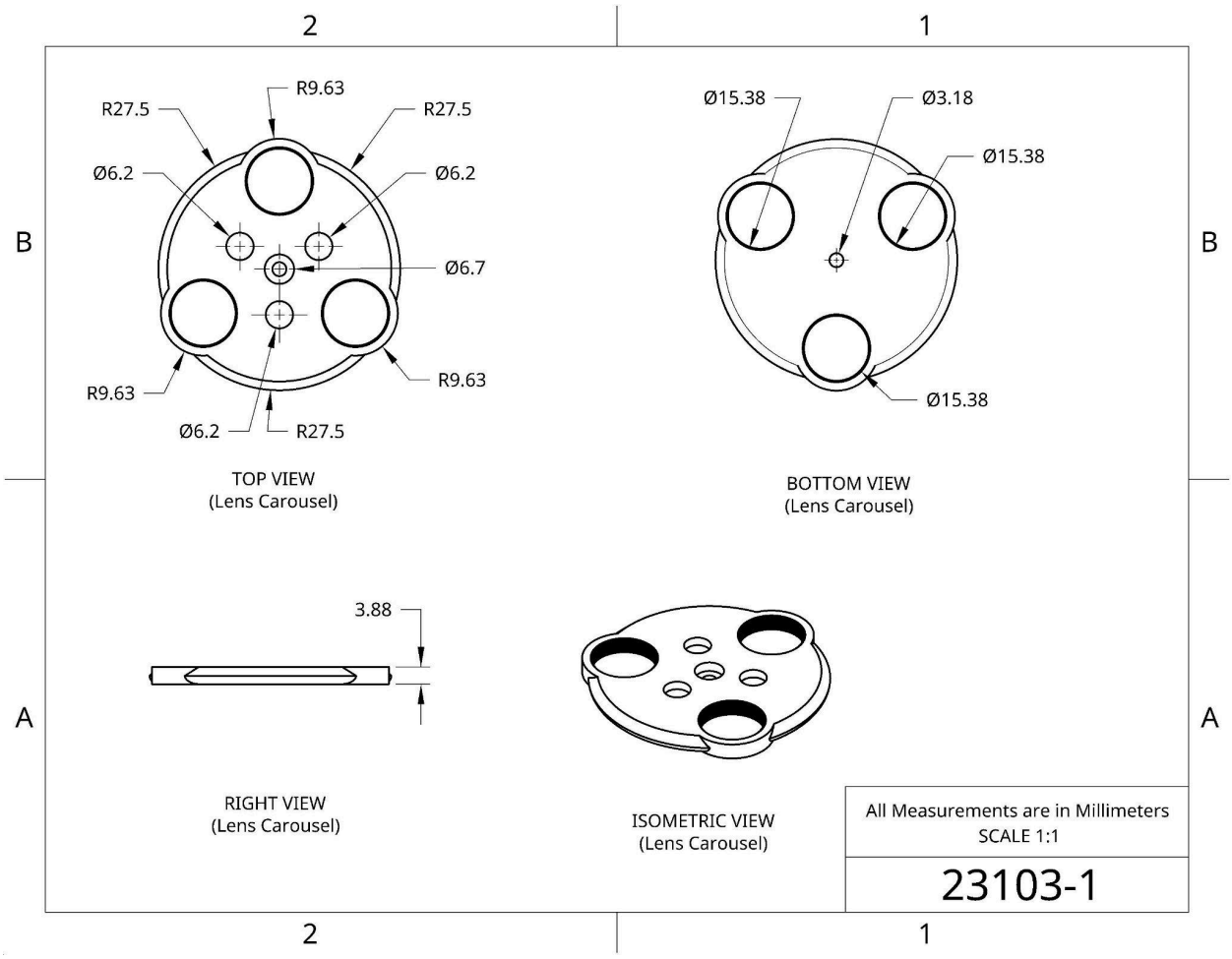
Note. The stage is a high-precision platform designed to hold standard microscope slides via two spring-loaded clips. Its geometry is optimized for the optical path, featuring an open center for transmitted illumination. Mechanically, the stage interfaces with the system through two distinct "jut-outs": a threaded connection for vertical actuation and a secondary clip that engages the square guide pillar. This constraint system effectively removes all degrees of freedom except for vertical travel, preventing the stage from tilting or shifting laterally. To minimize inertia and load on the focus threads, the stage is designed with minimal mass, utilizing strategic infill patterns to maintain high rigidity-to-weight ratios.

Figure 10: Stage Clips



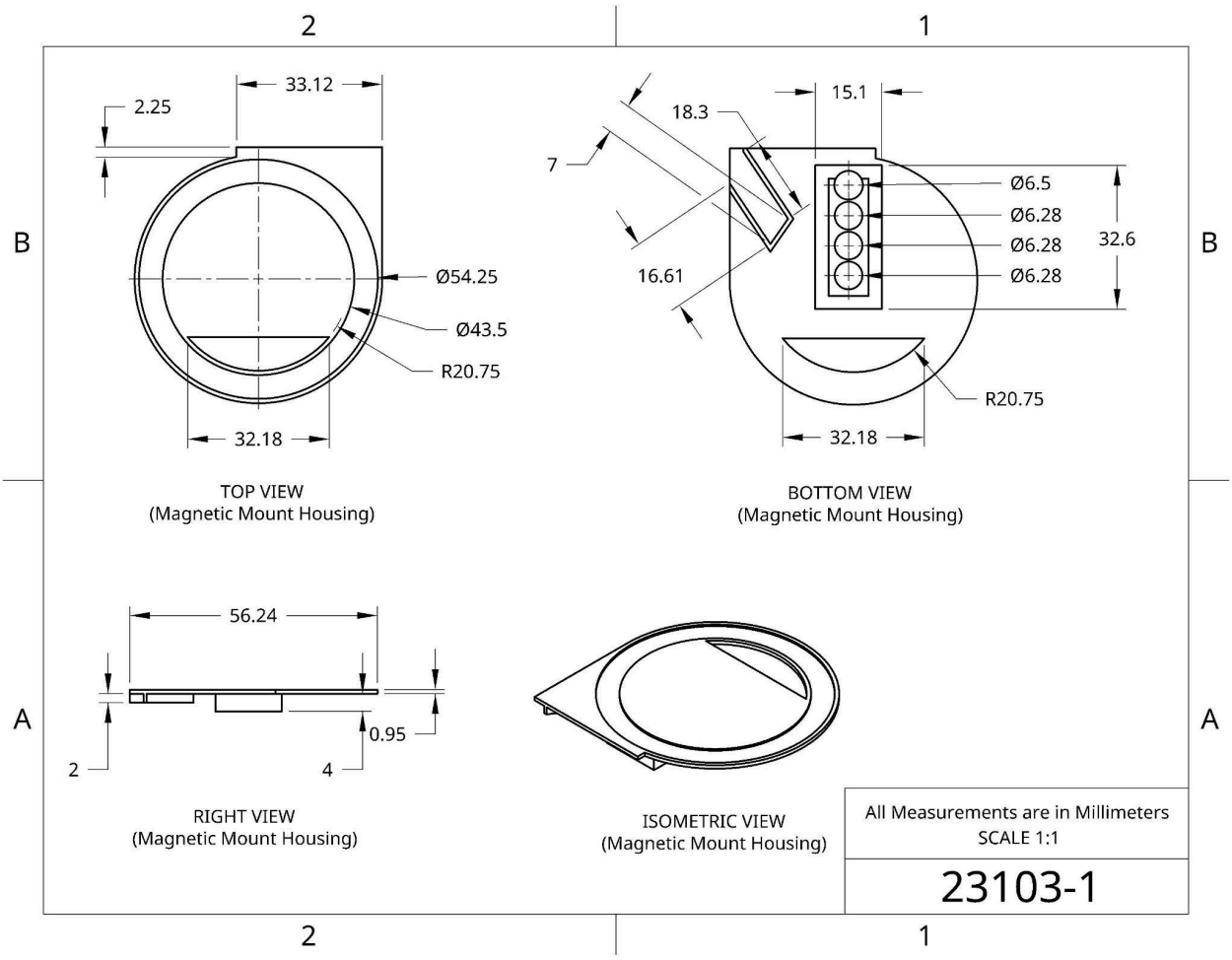
Note. The stage clips are precision-engineered spring retention elements designed to secure standard microscope slides flush against the stage platform. Fabricated as a continuous profile, each clip features a cantilevered arm that provides consistent clamping force while preventing lateral play or vertical deflection. The geometry of the hook tip is intentionally shallow to allow for intuitive, one-handed slide loading—a key usability requirement for educational settings. To preserve image quality, the clips are designed to contact only the extreme edges of the slide, ensuring the central optical path and the stage's open-center aperture remain completely unobstructed for transmitted illumination.

Figure 11: Lens Carousel



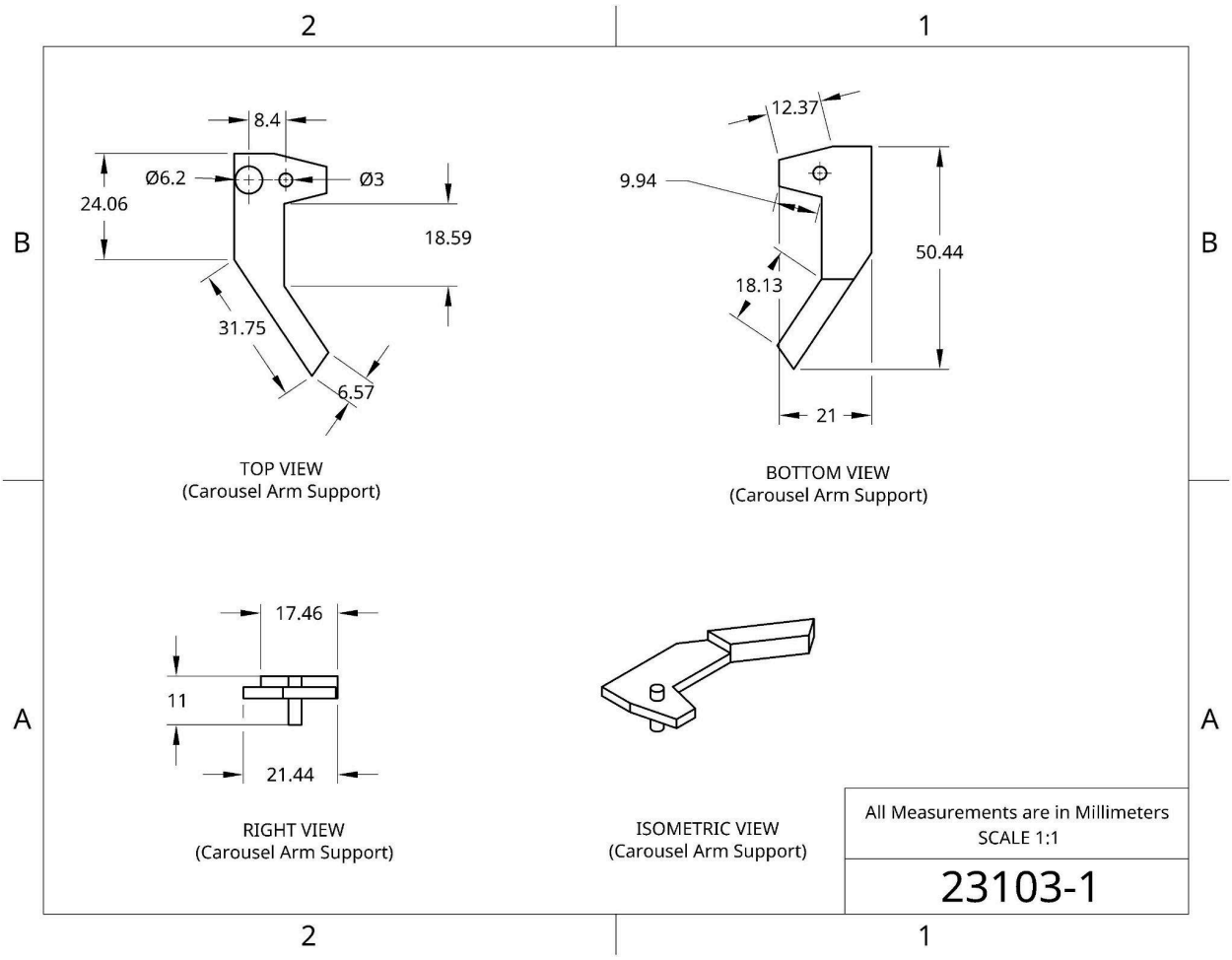
Note. The Lens Carousel is a rotating turret that supports three objective lenses (4x, 10x, and 40x). The design incorporates a central bearing to reduce rotational torque,, ensuring a smooth transition between magnifications. A key feature of this iteration is the magnetic "click-stop" system; magnets embedded in the bottom face align with the support arm to provide tactile feedback, guaranteeing the lens is perfectly centered over the camera sensor without requiring visual confirmation. The diameter was reduced by 30% from previous iterations to improve balance and reduce rotational inertia when mounted to a mobile device.

Figure 12: Magnetic Mount Housing



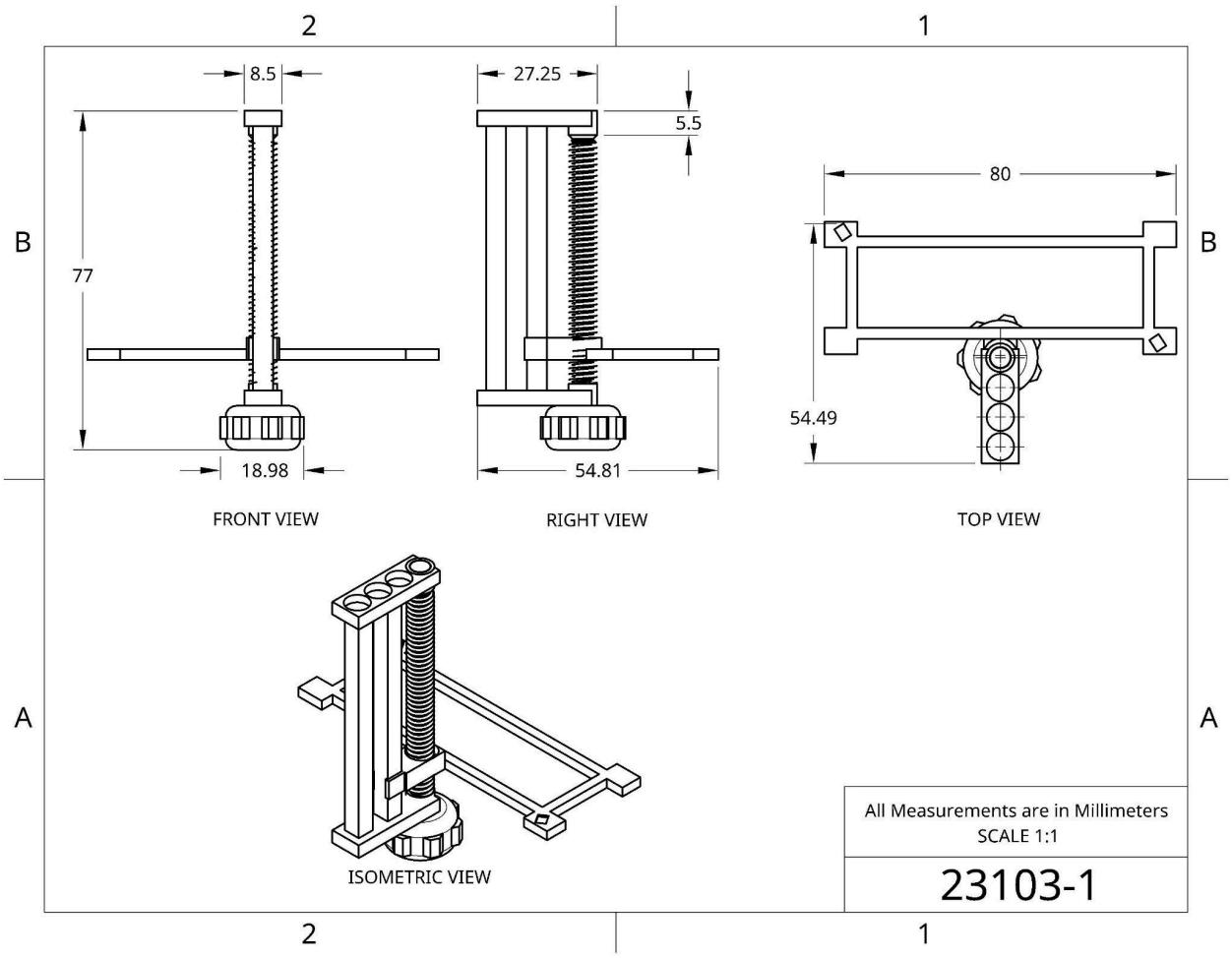
Note. Serving as the structural hub of the entire microscope, the magnetic mount housing facilitates the interface between the phone and the optical assembly. It features a precision-machined magnetic ring void for a press-fit magnetic attachment. To solve the issue of rotational alignment inherent in magnetic housing, an angular "jut-out" was added to register against the phone's camera bump, locking the housing into a single, repeatable position. Further refinements include a solid-fill center to eliminate flex under grip pressure and raised extrusions that guide the attachment of the carousel arm. This component acts as the geometric reference point; any deviation here would result in misalignment throughout the entire optical train.

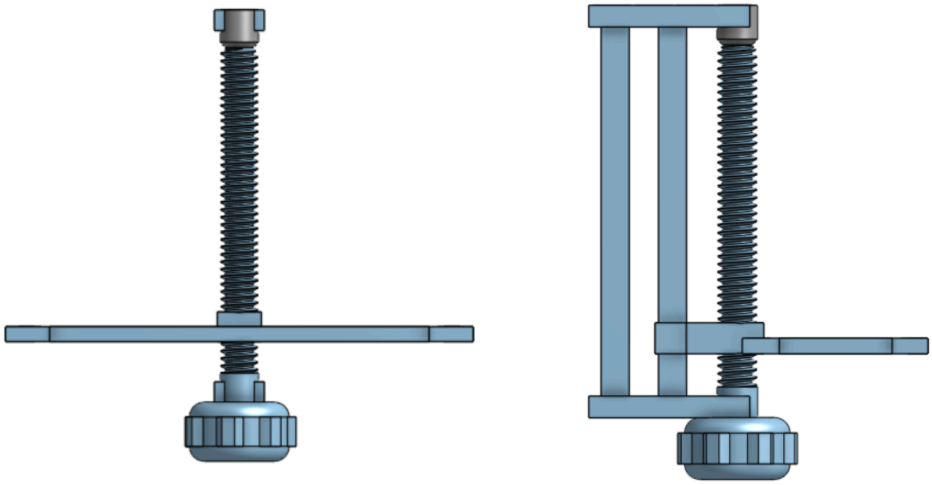
Figure 13: Carousel Arm Support



Note. The carousel arm support provides a rigid bridge between the magnetic housing and the lens turret. Its evolution was pivotal to the system's success: after earlier iterations where the arm obstructed lens rotation, the current design positions the arm on the left side, using the phone's edge as a vertical reference. This allows for an unobstructed optical path while positioning a magnet beneath the carousel to engage the click-stop mechanism. The arm is engineered for maximum stiffness to maintain a constant lens working distance, with a calibrated 2mm gap to optimize the magnetic attraction force for the indexing system.

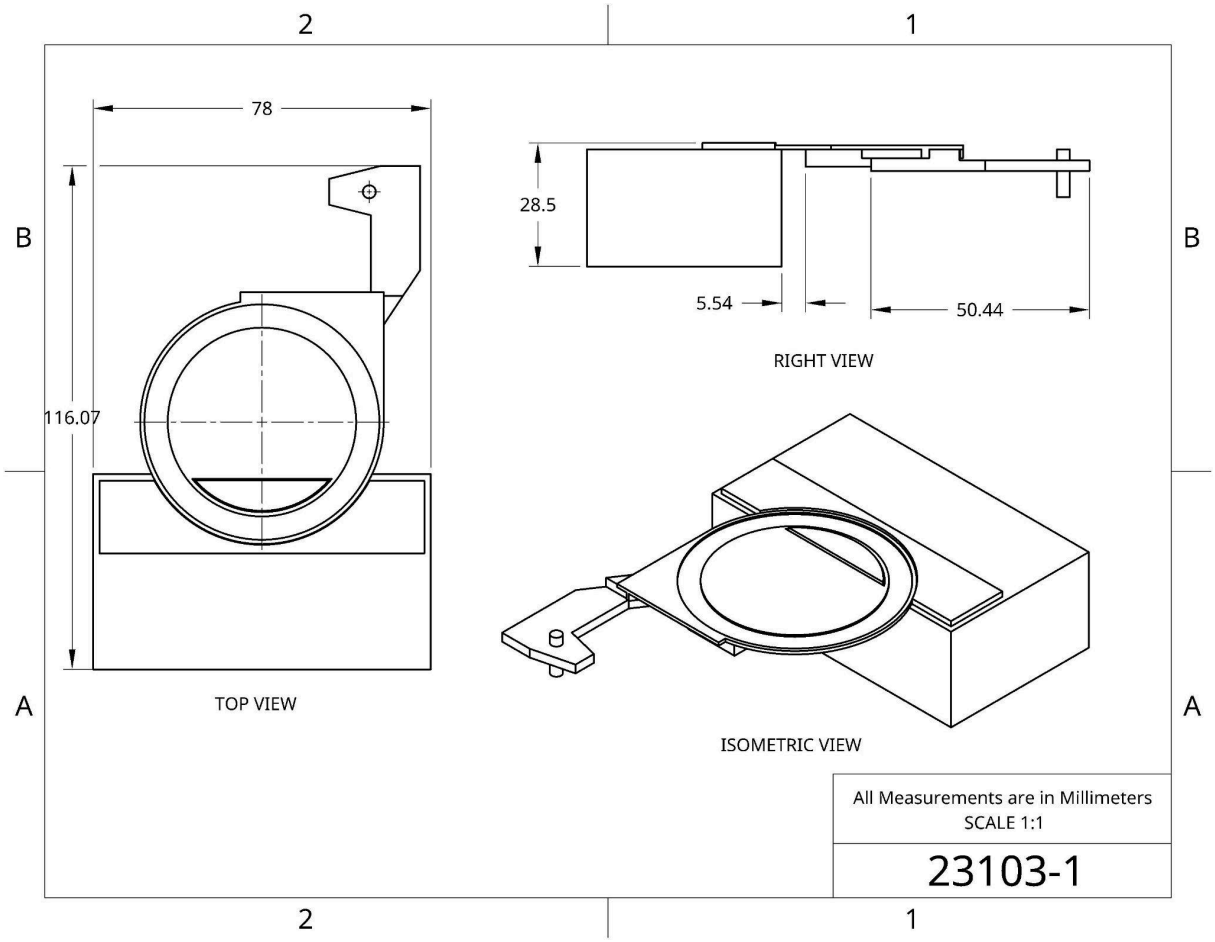
Figure 14: Vertical Arm, Knob, and Stage Assembly

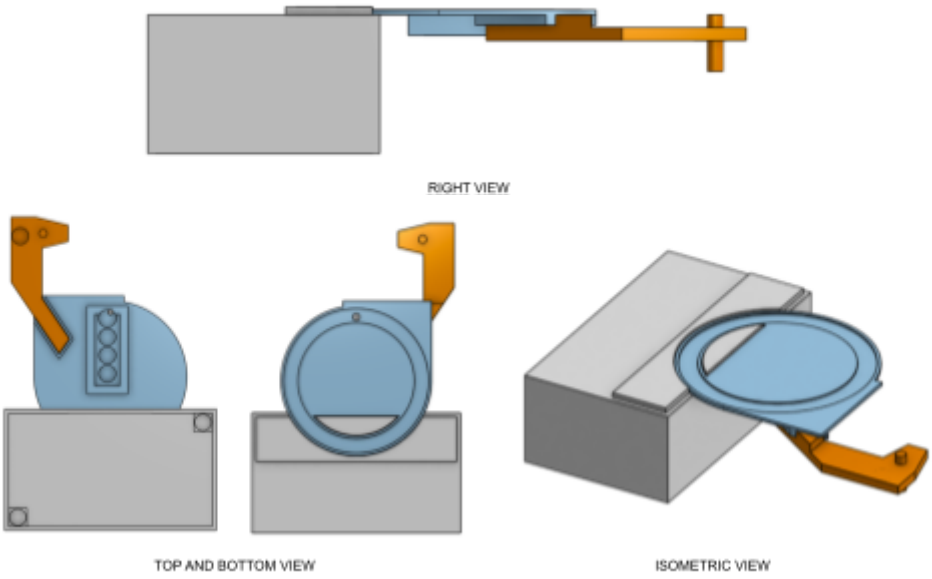




Note. This assembly integrates the vertical travel arm, focus knob, and specimen stage into a unified focus subsystem. The design utilizes a dual-pillar configuration, comprising a rear support pillar and a front square guide pillar to create a constrained channel for the stage. The threaded knob translates rotational input into precise vertical displacement, while the square guide pillar eliminates rotational drift induced by torque. This ensures the stage is restricted to pure axial motion, keeping the specimen centered on the optical axis. With a compact footprint of 77mm in height, the assembly is optimized for stable, phone-mounted operation.

Figure 15: Entire Phone Mount Module (Arm, Carousel, Magnetic Ring, Light Box) Assembly

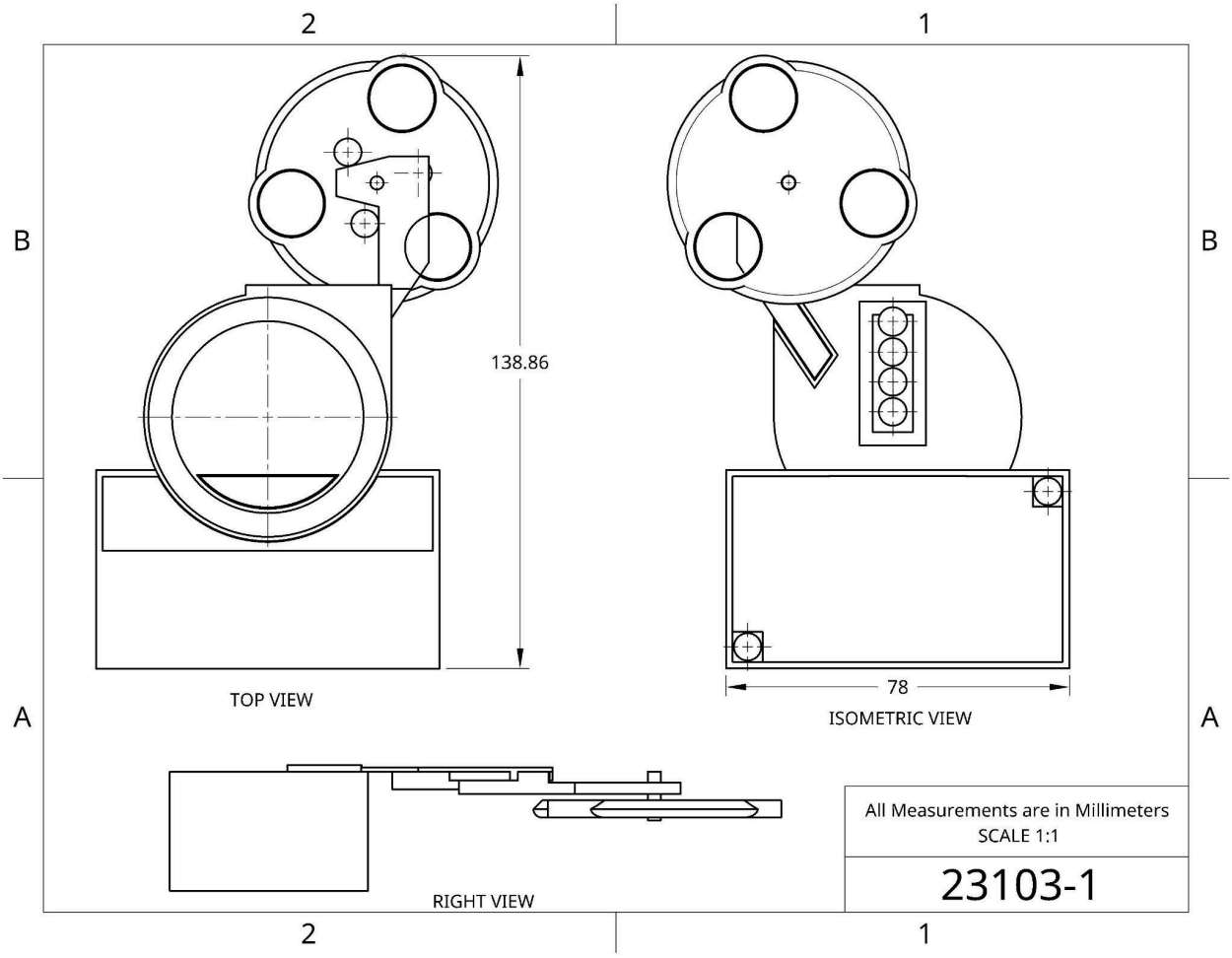




Note. The phone mount module serves as the fully integrated optical and alignment subsystem, combining the magnetic housing, carousel arm, lens carousel, and light box. The assembly uses the phone's edge as a vertical reference datum, while the magnetic ring and angular jut-out lock the entire optical train into a repeatable position against the camera bump. This module is engineered to maintain the specific working distance required for each objective lens. As the geometric reference point for the entire system, the magnetic housing ensures that all downstream optical elements remain perfectly aligned under handling pressure.

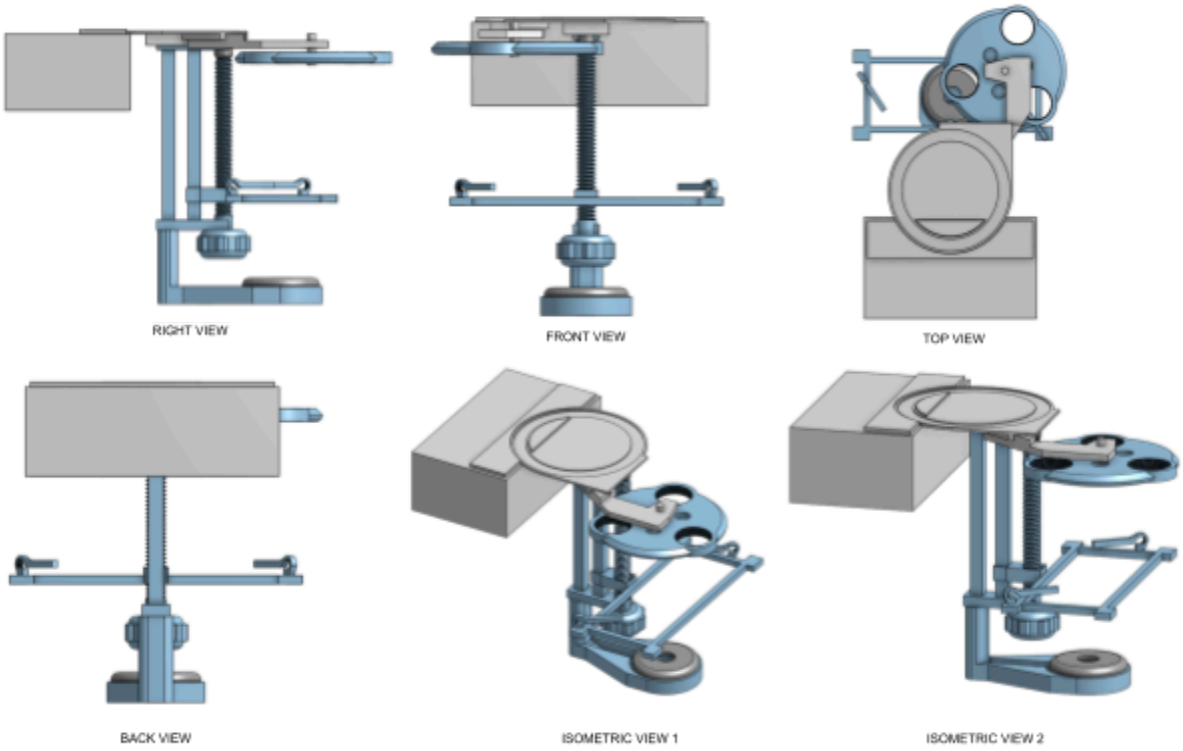
.

Figure 16: Microscope Assembly (Stage and Light Unattached)



Note. The light mounting setup provides the transmitted illumination path necessary for imaging biological specimens. Centered coaxially with the optical axis, the LED light source is positioned within a dedicated component box beneath the stage aperture. A diffuser layer is integrated between the LED and the specimen to redistribute the point-source output into a uniform illumination field, eliminating hotspots and improving image contrast. The modular design allows the light source to be detached or replaced independently, ensuring the illumination system remains serviceable without requiring a full disassembly of the microscope housing.

Figure 17: Full Microscope Assembly (Stage and Light Attached)



Note. The full microscope assembly represents the complete integration of all subsystems into a unified, functional instrument. With the stage and light source secured in their operational positions, the system establishes a continuous optical axis from the LED illumination source through the stage aperture, specimen slide, and objective lens to the phone camera sensor. The magnetic housing serves as the central geometric reference, ensuring all components maintain their calibrated working distances and alignment under normal handling conditions. This configuration demonstrates the modularity of the design, as each subsystem, the focus mechanism, lens carousel, and illumination unit can be independently accessed or serviced while the overall assembly remains structurally intact.

Microscopic Imaging Results and Observational Analysis

4× Magnification — Low Power Overview

At 4× magnification, the microscope provides a wide field of view, allowing two portions of the onion epidermis to be observed simultaneously. Individual cells are visible as repeating rectangular patterns, though fine internal structures are not clearly resolved. This magnification level is effective for initial sample positioning and general observation.

- Large field of view enables easy navigation of the specimen

- Cell boundaries are visible but lack fine detail
- Bright and stable image with minimal sensitivity to movement

Figure 18

Onion epidermis at 4× magnification



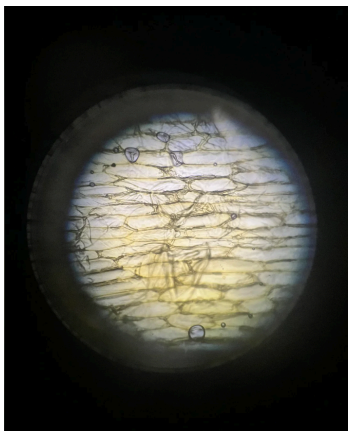
10× Magnification — Intermediate Detail

At 10× magnification, the clarity of the cell structure improves significantly. Cell walls become more defined, and the overall organization of the tissue is easier to interpret. This level provides a balance between field of view and detail, making it suitable for most standard observations.

- Improved definition of cell walls and structure
- Moderate field of view with increased detail
- Slightly reduced brightness compared to 4×
- Requires more precise focusing

Figure 19

Onion epidermis at 10× magnification



40× Magnification — High Detail Observation

At 40× magnification, the microscope reveals much finer structural detail within the onion epidermis. Individual cells fill a larger portion of the frame, and subtle variations in cell structure become more visible. However, this level also introduces greater sensitivity to alignment, lighting, and focus.

- High level of detail with clearly defined cell boundaries
- Very small field of view
- Reduced brightness due to increased magnification
- High sensitivity to vibration and focus adjustments

Figure 20

Onion epidermis at 40× magnification



Comparative Analysis of Magnification Levels

The imaging results demonstrate a clear trade-off between field of view and level of detail. Lower magnifications provide stability and ease of use, while higher magnifications enable more detailed observation but require greater precision in alignment and focusing. This progression reflects expected optical behavior and confirms that the system performs consistently across multiple magnification levels.

- 4× ⟶ Best for navigation and general observation
- 10× ⟶ Balanced performance for most use cases
- 40× ⟶ Best for detailed structural analysis

Overall, the results validate that the modular microscope is capable of producing meaningful and progressively detailed images, supporting both introductory and more advanced levels of scientific observation.

Design Constraints and Trade-offs

The development of the prototype required balancing several competing engineering constraints, each of which influenced the final design decisions. The most significant trade-offs include the following:

- **Cost vs. Optical Quality:**
    - Higher-quality optical components would improve image clarity but significantly increase cost, reducing accessibility.
- **Stability vs. Portability:**
    - A heavier structure improves stability but reduces ease of transport and usability in different environments.
- **Precision vs. Simplicity:**
    - More complex adjustment mechanisms increase control but make the system more difficult for beginners to use.

The final design represents a compromise that prioritizes accessibility, usability, and sufficient performance over maximum technical precision. This aligns with the project's goal of creating a practical educational tool rather than a laboratory-grade instrument.

System Limitations and Engineering Justification

While the prototype meets its primary design objectives, several limitations were identified during development. These include:

- Minor optical distortion due to low-cost lenses
- Sensitivity to precise alignment and focusing
- Reduced brightness at higher magnifications

However, these limitations are consistent with the design constraints and do not significantly impact the system's effectiveness for educational use. Importantly, they represent acceptable engineering trade-offs, given the emphasis on cost reduction and accessibility. Additionally, these limitations highlight opportunities for future improvement, such as higher-quality lenses or enhanced stabilization systems, reinforcing the modular nature of the design.

Engineering Design Process of the creation of the model:

Steps:	**Reasoning:**
Defining the problem	The initial step involved clearly identifying the core engineering challenge: the lack of accessible, affordable, and effective microscopy tools across both educational and real-world environments. While microscopes are essential for observing microscopic biological structures, materials, and environmental samples, traditional systems are often expensive, fragile, and not portable. This limits their use not only in classrooms, but also in field-based applications such as environmental science, on-site analysis, and remote research. The problem was therefore defined as designing a low-cost, durable, and smartphone-compatible microscope capable of producing clear and stable images in both controlled and field environments.
Understanding why this is a problem	This stage focused on analyzing the broader impact of limited access to microscopy tools. In educational settings, the lack of hands-on tools reduces engagement and limits the development of observational and analytical skills. In field science and real-world applications, the inability to perform quick, on-site microscopic analysis can delay data collection and reduce efficiency. Additionally, existing smartphone microscope attachments often fail due to poor alignment, instability, and inconsistent

	image quality, making them unreliable outside controlled conditions. Identifying these issues ensured that the design would address both usability and performance in diverse environments.
Potential solutions in the real world that already exist so we can build something unique	Existing solutions such as laboratory microscopes, portable field microscopes, and smartphone clip-on lenses were researched and evaluated. Traditional microscopes provide high-quality imaging but lack portability, are not practical for field use, and are extremely expensive. Portable microscopes improve mobility but are often still expensive or limited in functionality. Smartphone-based attachments are affordable but frequently suffer from alignment issues, instability, and poor optical consistency. This analysis revealed a gap: a solution that combines portability, affordability, and reliable performance. This led to the development of a modular system that improves alignment, stability, and usability while maintaining flexibility across multiple applications.
Planning our idea	The planning phase involved developing a system-level design that could function effectively in both controlled and variable environments. Key design requirements included portability, structural stability, optical clarity, and ease of use. The system was divided into subsystems (optical, structural, alignment, stage, and illumination), and initial sketches and CAD models were created to define component placement and interaction. Special emphasis was placed on ensuring that the system could maintain alignment and performance even under less controlled conditions, such as field use.
Prototyping our design	The prototype was constructed using accessible materials, including 3D-printed components, standard lenses, and LED lighting. The goal was to create a functional system that could be physically tested in real-world conditions. During assembly, attention was given to component fit, alignment accuracy, and overall structural rigidity. This stage allowed for the identification of practical challenges such as tolerance issues, stability limitations, and ease of setup, which are especially important for both classroom and field usability.
Testing our prototype	The prototype was tested under multiple conditions to evaluate its performance across different use cases. Optical testing was conducted using calibration methods and biological samples to assess magnification, clarity, and distortion. Stability testing evaluated how movement and vibration affected image quality, particularly in less controlled environments. Lighting tests were performed to determine effectiveness under varying illumination conditions. Additionally, usability testing assessed how easily users could set up, align, and operate the system without technical expertise. These tests ensured that the design was functional not only in ideal conditions, but also in practical, real-world scenarios.
Improving and iterating the design	Based on testing results, iterative improvements were made to enhance performance and reliability. Structural components were reinforced to reduce vibration and improve durability, especially for portable use. The alignment mechanism was refined to ensure consistent optical positioning with minimal user adjustment. The illumination system was optimized to provide more uniform lighting under different conditions. Optical performance was improved by refining lens placement and minimizing distortion. This iterative process ensured that the final design was not only functional but robust, adaptable, and suitable for both educational and field applications.

Full Iteration Process

I. Narrative Testing

The development of the modular smartphone microscope progressed through thirteen documented design iterations, each addressing specific engineering failures identified through physical testing, CAD evaluation, and optical analysis. The current prototype is a fully solid 3D-printed mockup produced in Onshape, designed to evaluate size, proportion, component geometry, and ergonomics. Because the model does not yet incorporate functional optics, a populated lens carousel, or a working LED illumination system, full empirical imaging performance testing cannot be conducted at this stage. Each component was evaluated individually throughout the iteration process rather than as a complete assembled system, which allowed specific failure points to be isolated and corrected before integration.

This testing approach reflects a controlled engineering methodology:

- Subsystems are validated independently before full system integration
- Failure modes are isolated to prevent compounded design errors
- Iterations are driven by observed performance limitations, not assumptions

Since a complete functional deployment requires assembled optics, a wired LED illumination system, and installed objective lenses, the following structured testing framework would be applied to the fully functional prototype to confirm the system meets its design objectives.

1. Optical Performance Testing

A printed calibration grid with 1mm spacing would be photographed at each carousel position under controlled LED illumination and analyzed for field of view, effective magnification, and edge distortion. Prepared biological slides, including onion epidermal cells and cheek epithelial cells, would serve as benchmarks to confirm the system resolves structures relevant to educational microscopy. Each carousel position would be tested independently to verify that the magnetic click-stop mechanism correctly places each objective at its required working distance from the specimen. Images would be evaluated on sharpness, brightness, and resolution using both visual inspection and basic image analysis software. Key evaluation criteria include:

- Magnification consistency across all objective lenses
- High-resolution quality that allows for the observation of biological structures
- Optical distortion (edge vs. center clarity)
- Repeatability of focal plane positioning

2. Structural and Stability Testing

At higher magnifications, even a very small physical displacement produces a visible image shift. The assembled prototype would be subjected to controlled disturbance testing, applying light lateral and vertical forces to the housing while observing the live camera feed for image shift or blur. The support leg would be evaluated to confirm

it prevents tipping under the phone's weight without introducing vibration into the optical path. Strategically placed infill patterns at structural joints would be assessed for their ability to dampen forces applied during carousel rotation and slide adjustment. Performance indicators include:

- Image displacement relative to applied force
- Structural rigidity under dynamic interaction
- Vibration damping effectiveness of printed joints
- Stability of the center of gravity during use

3. Magnetic Alignment and Repeatability Testing

A cross-hair target slide would be imaged after ten consecutive attach-detach cycles. Cross-hair position in each image would be compared to assess positional repeatability. Any meaningful variance between attachments would indicate the angular alignment, jut-out or camera bump registration requires further refinement. Measured variables include:

- Positional deviation between trials
- Angular misalignment across repeated attachments
- Consistency of optical axis alignment without user adjustment

4. Carousel and Illumination Testing

The bearing-supported carousel would be rotated through all three positions repeatedly to confirm smooth rotation and consistent magnetic snap engagement at each lens position. The LED illumination system would be evaluated by photographing a blank white slide and analyzing brightness uniformity across the field of view. Any visible center-to-edge brightness gradient would indicate the diffuser requires repositioning or redesign. The detachable light system would also be tested for connection reliability and ease of reattachment to confirm the modular design functions as intended. Evaluation includes:

- Rotational smoothness and torque consistency
- Precision of magnetic click-stop alignment
- Uniformity of illumination distribution
- Reliability of modular light system connections

5. Vertical Stage Focus Testing

The screw-lift stage mechanism would be evaluated by rotating the focus knob incrementally and observing the resulting vertical movement of the stage. The system must allow fine vertical adjustments without sudden or jerky movement, and the square guide pillar must prevent rotational drift during knob actuation. The sleeve at the top clip would be confirmed to spin freely, enabling smooth knob rotation without binding. Stage clips would be tested to confirm they securely hold a standard microscope slide without introducing play or lateral shift during adjustment. Critical focus metrics:

- Precision of vertical displacement per rotation
- Absence of rotational drift during adjustment
- Smoothness of motion (no binding or slipping)
- Stability of specimen positioning

6. Usability Testing

Five to ten teachers with varying scientific backgrounds would be asked to attach the device via magnetic, rotate the carousel to a selected lens, place a prepared slide on the stage, adjust focus using the screw knob, and capture an image, following only a brief written instruction card. Time to first focused image and number of errors would be recorded. Participants would then complete a Likert-scale survey rating ease of use. A successful result is defined as achieving focus within three minutes without assistance and a rating at or above 4 out of 5, at a total system cost below $25. Measured usability outcomes:

- Time to first successful image capture
- Number of user errors during setup
- Ease-of-use rating (Likert scale)
- Independence from prior technical experience

II. Refinements of the Prototype

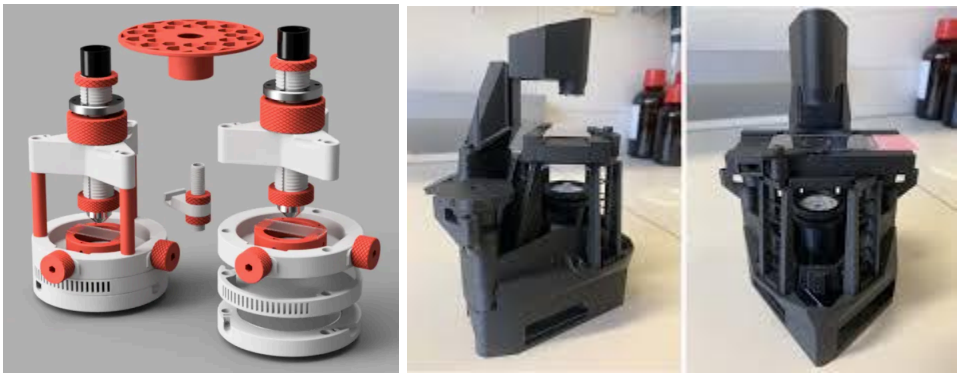
The following documents each major design iteration, the specific problem it addressed, and the engineering change implemented in response. The refinement process followed a consistent engineering framework:

- Identification of a specific failure mode
- Isolation of the root cause
- Implementation of a targeted design modification
- Validation through testing or modeling

Iteration 1 — Initial Reference Research: Galileo Microscope Blender Model

Before beginning CAD modeling, the team researched existing microscope designs and identified the Galileo microscope as a reference. A 3D render was produced in Blender to explore the geometry and optical layout. However, evaluation of this render revealed that a Galileo-style form factor was not compatible with the project's goals. The design was too large to be portable, removed the phone-mounted concept entirely, and would have required a fixed stand structure that would have reduced accessibility. The Galileo model was eliminated as a direct design reference, but the process of modeling and evaluating it confirmed the core requirement: the design must be compact, phone-integrated, and modular.

Figure 21 and 22



Iteration 2 — Phone Clip-Style Mount

The team pivoted to a phone clip-style mount that would physically clip to the back of the phone and position the lens over the camera. This was designed and printed for evaluation. Physical testing revealed two critical problems: the clip was bulky and added significant mass, and it was difficult to center precisely on the phone's camera. Because the camera must be optically aligned with the external lens consistently across every use, a clip that could shift position between attachments was fundamentally incompatible with the reliability requirement. The clip-mount approach was eliminated.

Iteration 3 — Magnetic Ring Housing

Research into magnetic technology revealed that virtually every recent iPhone includes a built-in magnetic alignment ring, making it a universally available attachment point. A housing was designed around an insertable magnetic ring that could magnetically attach to the back of the phone. This was printed and tested. The magnetic connection was strong and repeatable, resolving the centering and bulk issues of the clip mount. However, because the megnetic ring is a perfect circle, the attachment could rotate slightly between connections, introducing angular misalignment between the optical axis and the camera sensor.

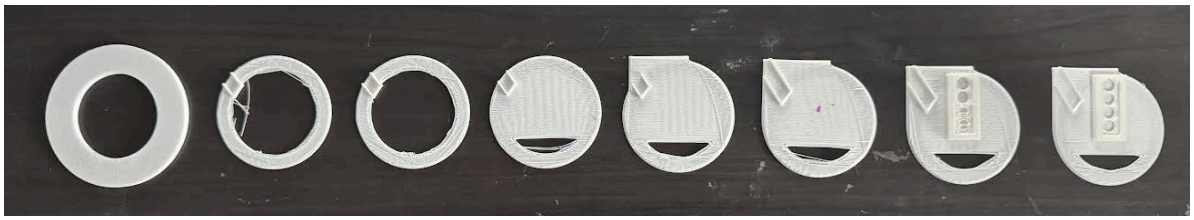
Figure 23



Iteration 4 — Magnetic Alignment Indent and Solid Ring Housing

To solve the rotational variance problem, two changes were made. First, the void in the center of the magnetic ring housing was filled solid, making the housing more rigid and less prone to flex under handling forces. Second, an angular jut-out was added to the upper-left portion of the magnetic ring housing. This jut-out registers against the phone's camera bump on every attachment, ensuring that the optical axis aligns with the camera consistently, regardless of how the user places the device. This converted alignment from a user-dependent variable into a mechanically guaranteed result.

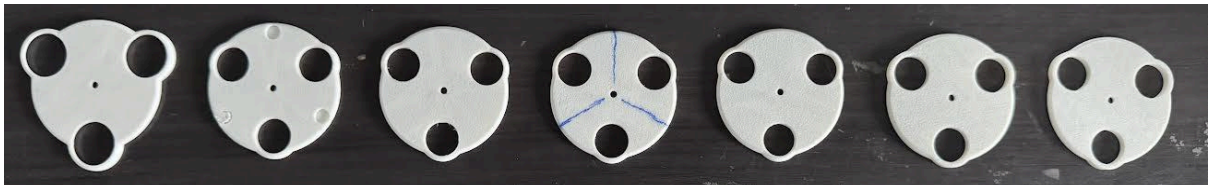
Figure 23



Iteration 5 — Initial Flat Lens Carousel

With the magnetic housing established, design began on the lens carousel. The first version was a flat circular disc with three holes that allowed lenses to be dropped or slid into place. Physical testing showed the lenses were not secure; they could shift within their holes under normal handling, displacing the optical axis and producing blurred images. This drop-in retention method was identified as a fundamental design flaw and was eliminated entirely.

Figure 24



Iteration 6 — Threaded Lens Seats and Magnetic Click-Stops

The carousel was redesigned with threaded lens seats, allowing each objective to be screwed firmly into position. This eliminated all lens displacement observed in Iteration 5. Additionally, three magnets were added to the top face of the carousel housing in front of each lens position. These magnets were designed to align with a corresponding magnet in the arm, allowing the carousel to snap into each lens position with a positive magnetic click-stop, ensuring the correct objective is always presented to the camera.

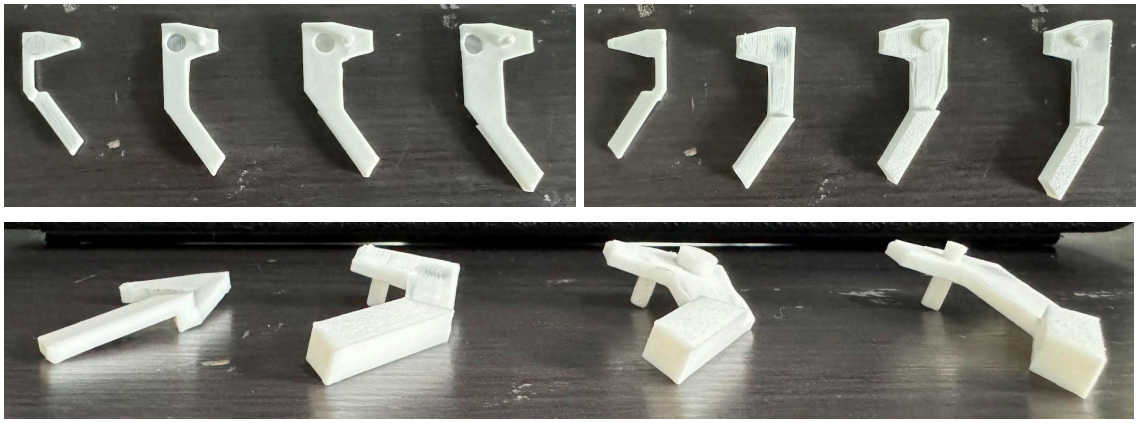
Figure 24



Iteration 7 — Arm Repositioning and Magnet Relocation

During rotation testing, it was discovered that the arm position directly below the carousel blocked lens rotation; the lenses physically interfered with the arm as the carousel turned. The magnets were relocated from the top face to the bottom of the carousel housing, and the arm was repositioned to the far left side of the device when viewed from the back. This placement took advantage of the angular support surface provided by the magnetic alignment jut-out, allowing the arm to use the phone's edge as a perfect vertical reference line. A corresponding magnet was integrated into the arm, enabling the carousel to snap precisely into each aligned position from below.

Figures 25. 26 and 27



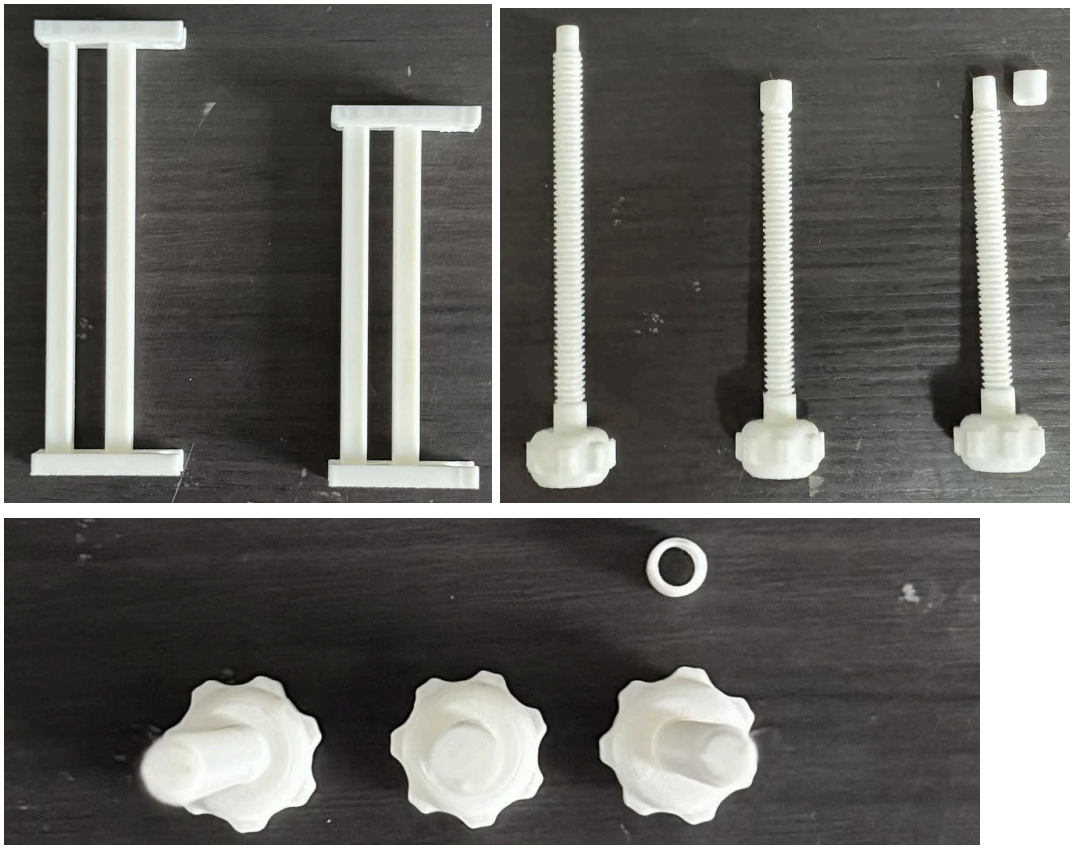
Iteration 8 — Carousel Size Reduction, Chamfering, and Bearing Integration

After repositioning the arm, the carousel geometry was redesigned to be significantly smaller in overall diameter, reducing mass and improving the balance of the attachment on the phone. Chamfered edges were added to both the top and bottom faces to improve grip, and small jut-outs were added around the perimeter as finger grips for spinning the carousel. During rotation testing of this version, substantial friction was observed at the central rotation point. A bearing was integrated into the center of the carousel to eliminate this friction, producing smooth, consistent rotation with minimal and predictable force at all three lens positions.

Iteration 9 — Vertical Stage Arm: Gear System to Screw-Lift Design

Design of the vertical stage arm began with a traditional microscope rack-and-pinion mechanism, a central gear aligned with a moving toothed bar connected to the stage. Physical evaluation revealed this approach was too bulky, too heavy, and geometrically incompatible with the compact phone-mounted form factor. The design was replaced with a screw-lift mechanism consisting of a rear main body support pillar, a square guide pillar in front of it, and two clips at the top and bottom to retain a threaded knob. The bottom clip was fixed as a solid connection. The top clip incorporated a sleeve that fit over the top of the knob and rotated freely, allowing the knob to spin in place smoothly without binding. This produced a compact, functional focus control mechanism.

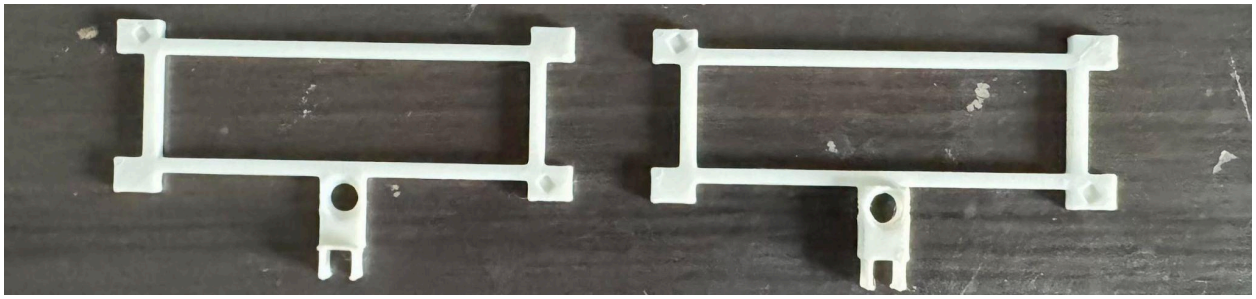
Figures 28, 29, and 30

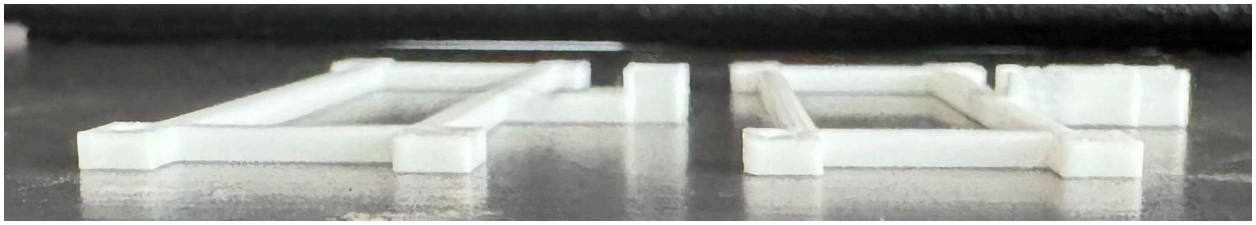


Iteration 10 — Specimen Stage Design

The stage was designed to be minimal and unobstructed. Two slide clips retain the specimen slide, and the center of the stage platform is entirely open to allow unobstructed light transmission from the LED below. A primary jut-out extending from the stage contains a threaded hole matching the thread pitch of the screw knob, allowing the knob's rotation to drive vertical stage travel. A secondary jut-out contains two arm-like clips that engage the square guide pillar, preventing any rotational movement caused by the knob and constraining the stage to pure vertical travel. This produced controlled, precise vertical motion in the most compact configuration achievable.

Figures 31 and 32





Iteration 11 — Stage-to-Housing Magnetic Connection and Alignment Features

Three magnets were added to the top of the vertical stage arm, and a corresponding raised square profile with three magnet indentations was added to the magnetic housing. This allowed the stage arm to attach magnetically to the housing with a defined and repeatable orientation, ensuring alignment between the stage, the lens carousel, and the camera on every assembly. The magnetic connection also keeps the stage removable, so the microscope can be used independently without slide-based samples. Custom 3D-printed stage clips were designed and integrated. Raised extrudes were also added to the magnetic housing to guide the lens carousel arm into precise alignment during attachment.

Figure 33



Iteration 12 — Light Box Design and Modular LED System

A housing for the illumination system was designed and connected directly to the magnetic housing. An LED was positioned beneath the stage opening to provide transmitted illumination. However, the initial light box housing was large and bulky, adding significant mass and making the overall device difficult to hold comfortably, one of the more significant ergonomic issues encountered across the entire iteration process. The light box was redesigned to be substantially smaller. The LED was made detachable from the light box, with the battery pack and connector cable remaining inside the housing, while the LED itself could be unplugged and removed independently. This preserved the modular design philosophy, and the illumination system can be separated from the main body without disassembling any other component.

Iteration 13 — Final Material and Color Selection

After completing all thirteen functional iterations, the team reviewed the accumulated physical prints and noted that the white PLA used throughout the iteration process produced prints that were difficult to photograph, visually uninspiring, and difficult to distinguish from one another across versions. The final iteration was printed in blue-gray PLA-colored filament, which improved visual contrast for documentation photography, gave the device a more professional appearance, and clearly distinguished the final version from all earlier iteration prints.

III. Reflection on Effectiveness

The modular smartphone microscope directly addresses the core problem identified at the outset of the project: the lack of an accessible, affordable, and reliable microscopy tool for educational environments. The thirteen-iteration development process, combined with the testing framework described in Section I, provides strong evidence of both the system's technical plausibility and its educational effectiveness.

Key indicators of effectiveness include:

- Progressive elimination of mechanical and optical failure modes
- Integration of engineering analysis (optics, alignment, stability)
- Transition from user-dependent to mechanically-controlled systems

The depth of the iteration process itself is meaningful evidence. Each of the thirteen iterations was driven by a specific observed failure: rotational misalignment, lens displacement, carousel friction, stage rotation, light box bulk, and each produced a targeted engineering solution that resolved that failure before the next phase of development began. This iterative structure mirrors professional engineering practice and demonstrates that the design has been progressively validated rather than assumed to work from the outset.

The camera lens selection process used the thin lens equation, the Rayleigh resolution criterion, depth of field analysis, and light collection efficiency comparisons to eliminate the ultrawide and telephoto lenses and confirm the main camera as the optimal imaging interface. This provides a documented, quantitative basis for one of the most consequential decisions in the system's optical design, demonstrating that engineering analysis rather than trial and error drove the selection.

The magnetic alignment system, refined through Iterations 3 and 4, is one of the most impactful design decisions in terms of usability. By using the angular jut-out to register against the camera bump, alignment is guaranteed by device geometry rather than user skill. This directly addresses one of the most common failure modes in existing smartphone microscope solutions, inconsistent optical axis alignment, and transforms it from a recurring problem into a solved constraint. The magnetic click-stop system developed in Iterations 6 and 7 similarly converts lens selection from a judgment call into a mechanically enforced outcome, ensuring the correct objective is always presented to the camera without requiring the user to visually confirm alignment.

The usability testing protocol would provide the most direct evidence of whether these design decisions translate to real-world accessibility. A successful result, most users achieving a focused image within three minutes without technical assistance, would confirm that the combination of magnetic alignment, threaded lens retention, bearing-supported rotation, screw-lift focus control, and modular stage design collectively produces a system that is genuinely operable by non-expert users. Taken together, the system demonstrates the following:

- Technical feasibility through validated engineering principles
- Functional reliability through iterative refinement
- Practical usability for real-world educational and field applications

IV. Other Issues Identified During the Iteration Process

Throughout the iteration process, several broader engineering challenges were identified:

- Scaling limitations of traditional mechanisms
- Manufacturing tolerances in FDM 3D printing
- Ergonomic constraints not visible in CAD
- Cross-device compatibility considerations

Gear System Incompatibility:

The initial rack-and-pinion focus mechanism explored in Iteration 9 was geometrically incompatible with the compact phone-mounted form factor. Traditional microscope focus mechanisms are designed for large, rigid bodies that can absorb the mechanical loads of a gear system. Scaling this down to a phone-attached accessory introduced unacceptable bulk and weight, highlighting the importance of not assuming that established microscope designs can be directly miniaturized, a compact, phone-integrated device requires purpose-designed mechanisms rather than scaled versions of conventional components.

FDM Printing Tolerance Accumulation:

Throughout the iteration process, dimensional tolerance in FDM-printed components became a recurring constraint. Standard PLA printing on consumer-grade printers carries tolerances that vary depending on geometry and printer calibration. For critical fit surfaces such as the threaded lens seats, magnetichousing fit, and stage clip geometry, accumulated tolerance across mating parts produced friction, play, or misalignment that required reprinting with adjusted dimensions. This will need to be managed systematically in the functional prototype through careful tolerancing at each interface surface.

Light Box Ergonomic Impact:

The initial light box design in Iteration 12 was large enough to meaningfully change the grip geometry of the assembled device, making it difficult to hold comfortably during slide adjustment and carousel rotation. This highlighted a design constraint that is difficult to evaluate in CAD alone, the ergonomic impact of adding a component can only be fully assessed by printing and physically handling the updated assembly. The issue was resolved by reducing the light box footprint, but it reinforced the importance of evaluating assembled ergonomics at each major design stage rather than only reviewing individual components in isolation.

Rotational Drift in Stage Travel:

During testing of the screw-lift focus mechanism, the rotation of the knob introduced a small rotational moment on the stage that, without the square guide pillar and dual-clip constraint system, caused the stage to drift rotationally rather than traveling purely vertically. This drift would shift the specimen laterally out of the field of view during focusing. The square pillar and clip system resolved this, but the issue illustrated how seemingly minor mechanical coupling effects become significant at the scale of a microscope system, where small displacements directly affect image quality.

Magnetic Cross-Model Compatibility:

The alignment system is calibrated specifically for the iPhone 15 Pro, using the camera bump position and magnetic ring geometry of that model as fixed references. While magnetic is present across multiple iPhone models, the camera position relative to the magnetic ring varies slightly between generations. This means the current alignment jut-out geometry may not provide correct optical axis alignment on models other than the iPhone 15 Pro

without modification. A family of model-specific jut-out inserts, or an adjustable alignment system, would be required to extend compatibility across all magnet-equipped devices and represents a clear direction for future development.

Communication of Solution:

The Modular Smartphone Microscope is an innovative solution that directly addresses the annual theme, "*Engineering the Tools of Scientific Discovery*," by transforming widely accessible consumer technology into a reliable, high-performance scientific instrument. By targeting the fundamental gap between the demand for microscopy tools in education and the practical limitations of traditional laboratory equipment, this system empowers students, educators, and field researchers to engage in meaningful scientific observation without the financial and logistical barriers associated with conventional microscopes.

The system integrates a magnetic-based optical alignment mechanism, a bearing-supported rotating lens carousel, a screw-lift vertical focus stage, and a modular LED illumination subsystem into a single compact, smartphone-compatible device. Together, these engineered subsystems replicate the core functionality of a traditional microscope at a fraction of the cost, while prioritizing portability, durability, and ease of use for non-expert users. Rather than relying on user skill to achieve alignment, the device uses mechanical constraints and magnetic registration to guarantee repeatable optical performance on every attachment, converting one of the most common failure points in smartphone microscopy into a solved engineering constraint.

Traditional microscopy tools advance science but remain confined to well-funded laboratories, leaving the majority of students without direct access to one of science's most foundational instruments. The Modular Smartphone Microscope redefines what constitutes a scientific tool by engineering a system that meets the same observational objectives at a significantly lower cost and broader scale, directly embodying the theme of engineering new tools to expand access to scientific discovery.

To summarize, the Modular Smartphone Microscope addresses the theme of Engineering the Tools of Scientific Discovery through several key mechanisms:

- The magnetic alignment jut-out registers against the phone's camera bump, guaranteeing optical axis alignment on every attachment without user calibration.
- The magnetic click-stop carousel presents three interchangeable objective lenses at precisely repeatable working distances, enabling consistent magnification across all users and sessions.
- The screw-lift focus stage with square guide pillar constrains specimen travel to pure vertical motion, eliminating rotational drift and enabling fine, controlled focusing for users with no prior microscopy experience.
- The modular LED illumination system provides uniform transmitted lighting through a diffuser layer, directly improving image contrast and resolution at the smartphone sensor.
- The fully 3D-printed, PLA-based construction maintains a target system cost below $25, enabling scalable reproduction in classrooms, makerspaces, and low-resource communities worldwide.

The design prioritizes modularity and long-term adaptability, allowing individual subsystems to be upgraded or replaced independently as the project continues to develop. By shifting microscopy from a passive, demonstration-based classroom experience to an active, hands-on investigation accessible to every student, the Modular Smartphone Microscope fulfills the core engineering goal at the heart of this year's theme: designing the tools that make scientific discovery possible.

References

Anatomy of a Microscope | Microscope Anatomy Guide | Evident. (2026). Evidentscientific.Com. https://evidentscientific.com/en/microscope-resource/knowledge-hub/anatomy/introduction

Breslauer, D. N., Maamari, R. N., Scwitz, N. A., Lam, W. A., & Fletcher, D. A. (2009). Mobile Phone Based Clinical Microscopy for Global Health Applications. *PLoS ONE*, *4*(7), e6320. https://doi.org/10.1371/journal.pone.0006320

Courtney, A., Alvey, L. M., George, Burke, N., & Pickering, M. (2020). The Flexiscope: a low cost, flexible, convertible and modular microscope with automated scanning and micromanipulation. *Royal Society Open Science*, *7*(3), 191949–191949. https://doi.org/10.1098/rsos.191949

Deep learning transforms smartphone microscopes into laboratory-grade devices. (2018). Ucla.Edu. https://samueli.ucla.edu/deep-learning-transforms-smartphone-microscopes-into-laboratory-grade-devices/

Dimensions.com. (2020). Dimensions.Com. https://www.dimensions.com/element/apple-iphone-15-pro-17th-gen

Donelan, A. W., Lemer, H. N., Roy, S., & Butzin, N. C. (2025). 3D Printed Microscopes as a Low-Cost Alternative to Conventional Light Microscopes. *ILearn Teaching Resources*. https://doi.org/10.62812/xlce8013

Dri, C., Mirco Panighel, Tiemann, D., Patera, L. L., Troiano, G., Yves Fukamori, Knoller, F., Lechner, B. A. J., Cautero, G., Giuressi, D., Comelli, G., Jordi Fraxedas, Africh, C., & Esch, F. (2019). The new FAST module: A portable and transparent

add-on module for time-resolved investigations with commercial scanning probe microscopes. *Ultramicroscopy*, *205*, 49–56. https://doi.org/10.1016/j.ultramic.2019.05.010

Lightley, J., Kumar, S., Lim, M. Q., Garcia, E., Görlitz, F., Alexandrov, Y., Parrado, T., Hollick, C., Steele, E., Roßmann, K., Graham, J., Broichhagen, J., McNeish, I. A., Roufosse, C. A., Neil, M. A. A., Dunsby, C., & French, P. M. W. (2023). *openFrame*: A modular, sustainable, open microscopy platform with single‐shot, dual‐axis optical autofocus module providing high precision and long range of operation. *Journal of Microscopy*, *292*(2), 64–77. https://doi.org/10.1111/jmi.13219

Lu, Q., Liu, G., Xiao, C., Hu, C., Zhang, S., Xu, R. X., Chu, K., Xu, Q., & Smith, Z. J. (2018). A modular, open-source, slide-scanning microscope for diagnostic applications in resource-constrained settings. *PLoS ONE*, *13*(3), e0194063–e0194063. https://doi.org/10.1371/journal.pone.0194063

Microscope Parts & Specifications Labeled Diagram | Microscope World Resources. (2026). Microscopeworld.Com. https://www.microscopeworld.com/microscope-parts-and-specifications/

Parts of a Microscope with Functions and Labeled Diagram. Microbe Notes. https://microbenotes.com/parts-of-a-microscope/

Review of Apple 25W MagSafe Charger. (2025, January 28). Chargerlab. https://www.chargerlab.com/review-of-apple-25w-magsafe-charger/

Roney, J. A. (2025, June). Engineering Tools of Scientific Discovery in Popular Culture, Part I in a Series of Thematic Courses Introducing Non-Majors to Space Exploration Concepts and Topics. In 2025 ASEE Annual Conference & Exposition.

Rühl, H. (2012, May 2). *Optical Microscopes – Some Basics*. Leica-Microsystems.Com. https://www.leica-microsystems.com/science-lab/microscopy-basics/optical-microscopes-some-basics/

Switz, N. A., D'Ambrosio, M. V., & Fletcher, D. A. (2014). Low-Cost Mobile Phone Microscopy with a Reversed Mobile Phone Camera Lens. *PLoS ONE*, *9*(5), e95330. https://doi.org/10.1371/journal.pone.0095330

Testing and Detection Application Notes. (2024). Edmundoptics.Com. https://www.edmundoptics.com/knowledge-center/application-notes/testing-and-detection/

The basics of optical alignment - 2024 - Wiley Analytical Science. (2024). *Analytical Science Article DO Series*. https://doi.org/10.1002/was.0004000445

The OpenFlexure Project. (2026). *OpenFlexure Microscope - OpenFlexure*. The OpenFlexure Project. https://openflexure.org/

World, M. (2018, May 3). *Tips for Optimizing Microscope Illumination*. Microscope World. https://www.microscopeworld.com/blog/tips-for-optimizing-microscope-illumination/

World, M. (2023, August 13). *Functions of Fine and Coarse Adjustment Knobs in Microscopes*. Microscope World. https://www.microscopeworld.com/blog/functions-of-fine-and-coarse-adjustment-knobs-in-microscopes/